COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4

Module 2
Routing and Addressing at Internet Scale

1 · Inter-AS routing and AS policy | 2 · Intra-AS routing protocols | 3 · Aggregation, NAT and IPv6

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RECAP · MODULE 1

A quick recap of routing protocols

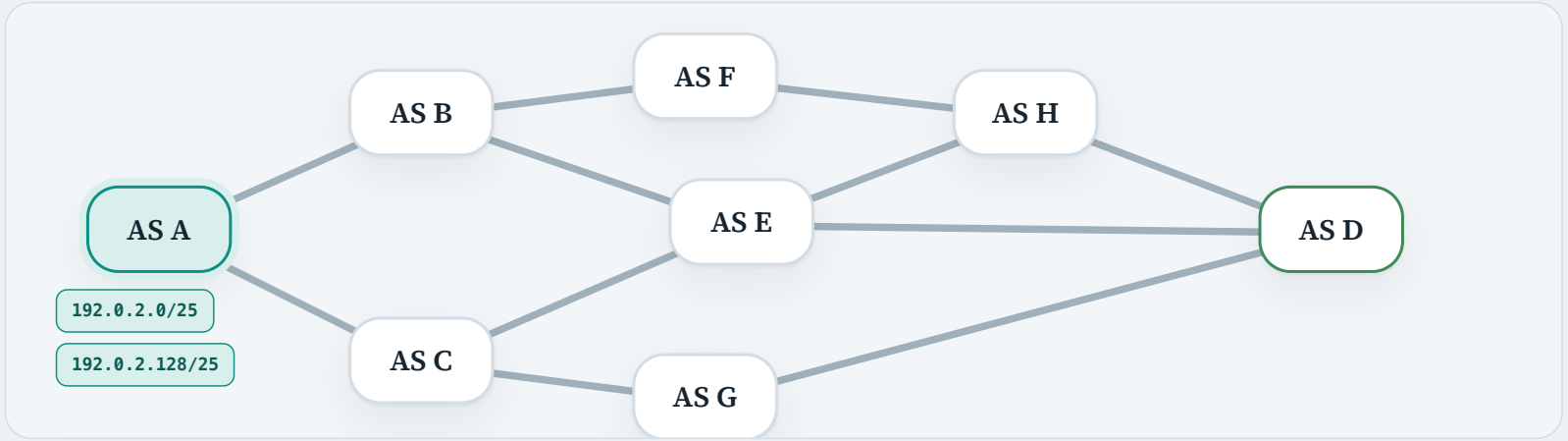
We can categorise a routing protocol by the **algorithm** underneath it — in Module 1 we met **distance vector** and **link state**.

We can also categorise it by **where it runs** — inside one autonomous system, or between them.

algorithm ↓ / domain →	**Intra-AS** within one autonomous system	**Inter-AS** between autonomous systems
distance vector	**RIP** Routing Information Protocol next	
link state	**OSPF** Open Shortest Path First next	
path vector a close relative of distance vector		**BGP** Border Gateway Protocol now

THE INTER-AS ROUTING PROBLEM

The Internet is a network of autonomous systems



INSIDE AN AS

- An AS contains **many routers**, which together form **several subnets**, all managed by a single organisation.
- Each subnet is identified by a **network prefix**, written in **CIDR** notation — AS A holds `192.0.2.0/25` and `192.0.2.128/25`.
- So `192.0.2.1` sending to `192.0.2.200` stays **inside AS A**: that is **intra-AS routing**.

AS and network-prefix concepts: RFC 4271 §§1, 3

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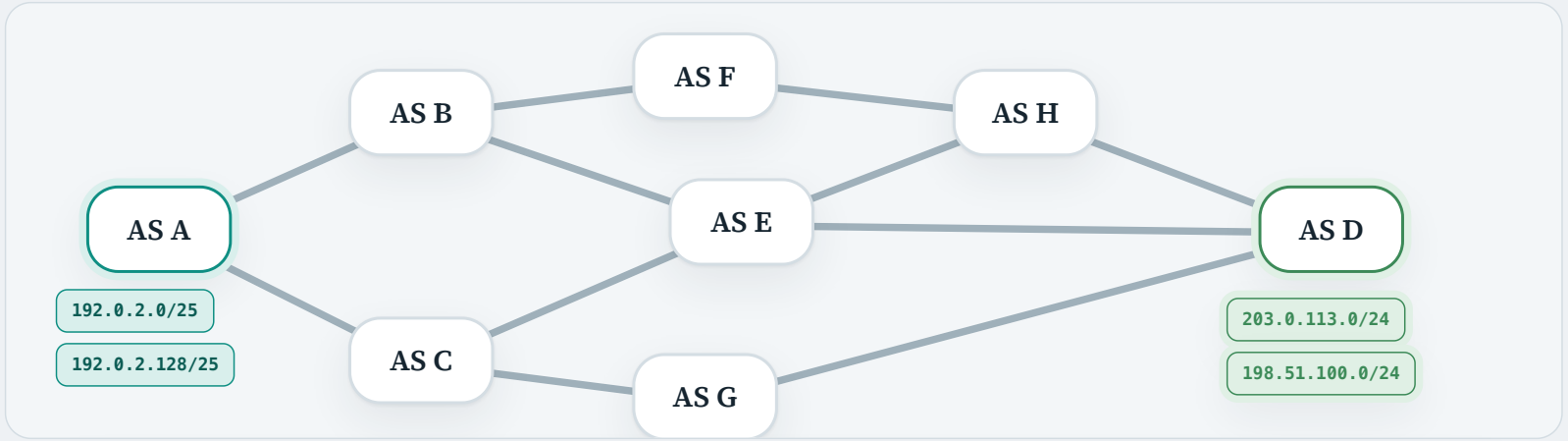


INTER-AS ROUTING PROBLEM

Now `192.0.2.1` sends instead to `203.0.113.42` — an address in **AS D**.

AS and network-prefix concepts: RFC 4271 §§1, 3

A SYSTEMATIC SOLUTION TO THE INTER-AS PROBLEM

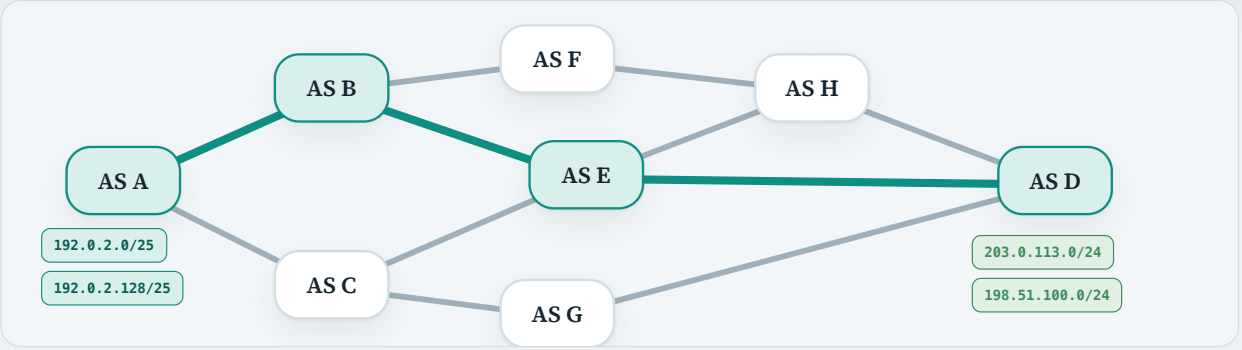
BGP is the Internet's inter-AS routing protocol

BGP is the Internet's answer to that problem. Here is how it does it.

LEARN EXTERNAL REACHABILITY

- Route-finding over a graph of **nodes** and **edges**, just as in Module 1 — except a node is no longer a router but a whole **autonomous system**, holding one or more subnets.
- Unlike Module 1, a destination is not a single IP address. It is a **destination prefix** in CIDR notation, like `203.0.113.0/24`, identifying a subnet inside some AS.
- For every **source–destination pair**, find a path across that graph — a path made of several ASes.

e.g. a path from AS A to AS D: **AS A → AS B → AS E → AS D**



BGP purpose and operation: RFC 4271 §§1, 3, 9.1

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DISTRIBUTE IT INSIDE THE AS

Once those external paths are known, the AS's own routers must be told, so they know how to forward packets leaving the AS.

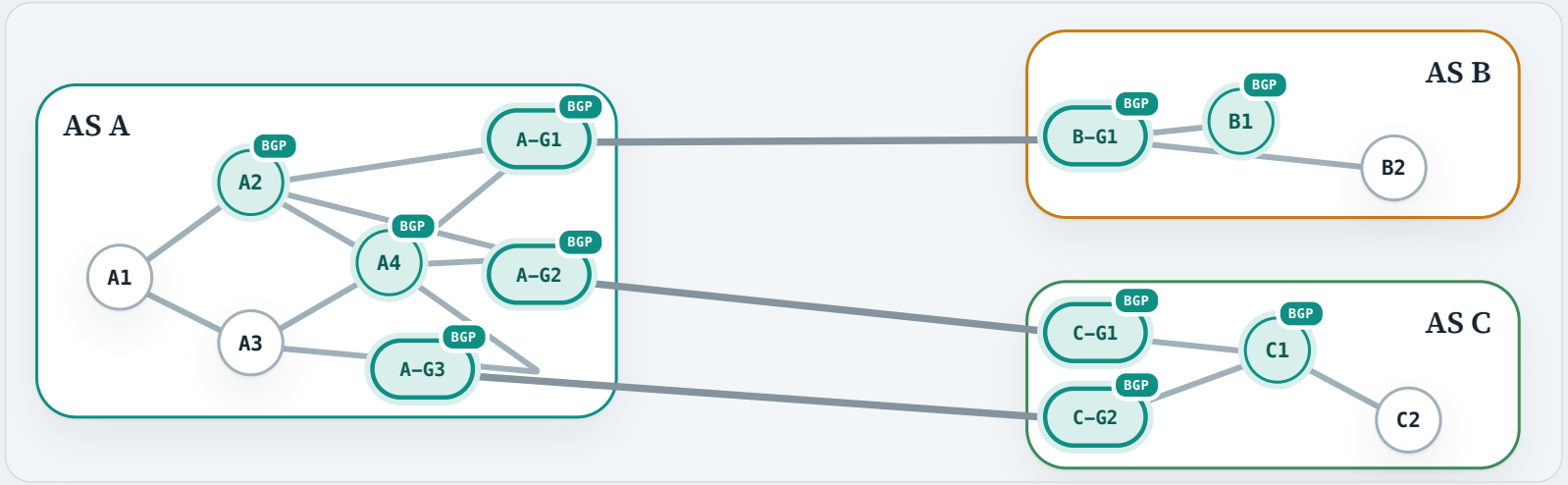
SELECT A PREFERRED ROUTE

Again several paths may reach the same destination. In Module 1 the criterion was simple — the **shortest** path, one least cost. Here it is **multidimensional**: economic incentives, delay, and more. BGP supplies the criteria that pick the best one.

BGP purpose and operation: RFC 4271 §§1, 3, 9.1

WHO PARTICIPATES IN BGP?

BGP is run by BGP-speaking routers



BGP SPEAKER

A router that runs BGP and maintains configured BGP sessions with selected peer routers. Each AS decides which routers will participate.

BGP speaker definition: RFC 4271 §1.1

WHO PARTICIPATES IN BGP?

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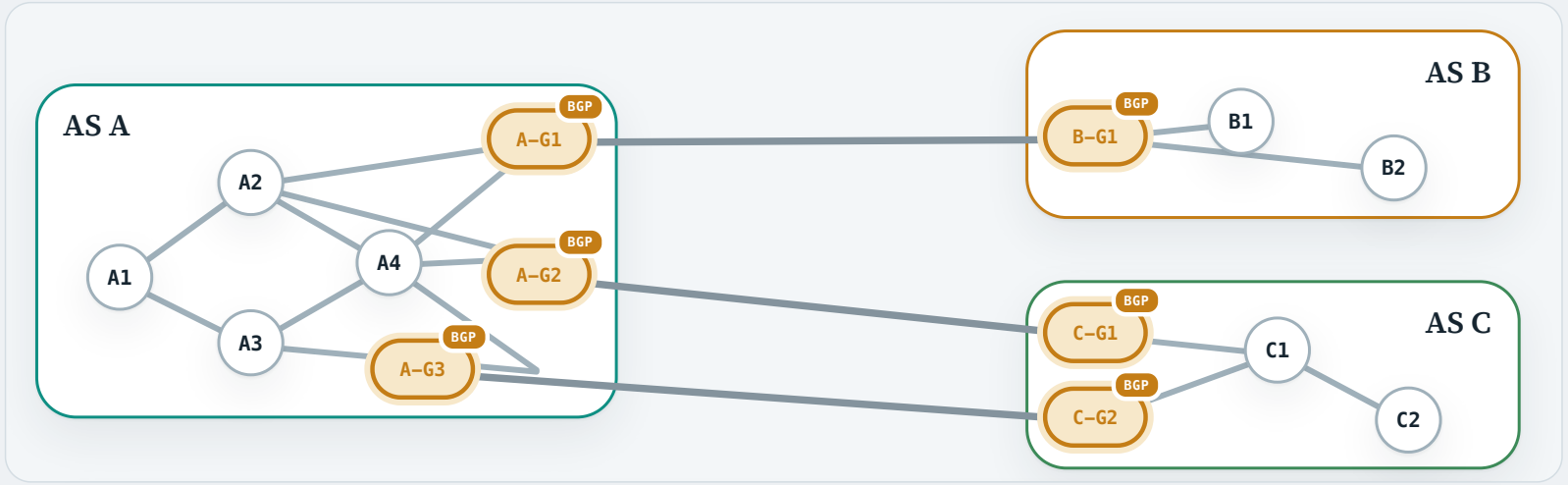

USUALLY AT THE EDGE

A border or gateway router links directly to a router in another AS. Such routers normally run BGP; here, all six gateways are BGP speakers.

BGP speaker definition: RFC 4271 §1.1

WHO PARTICIPATES IN BGP?

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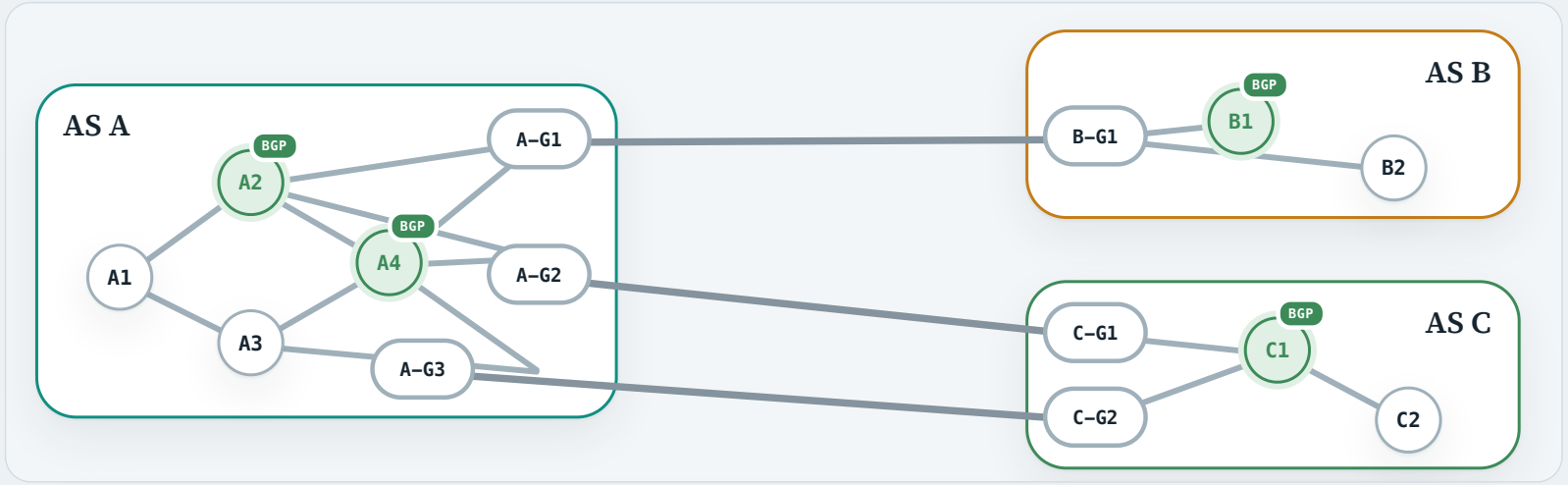


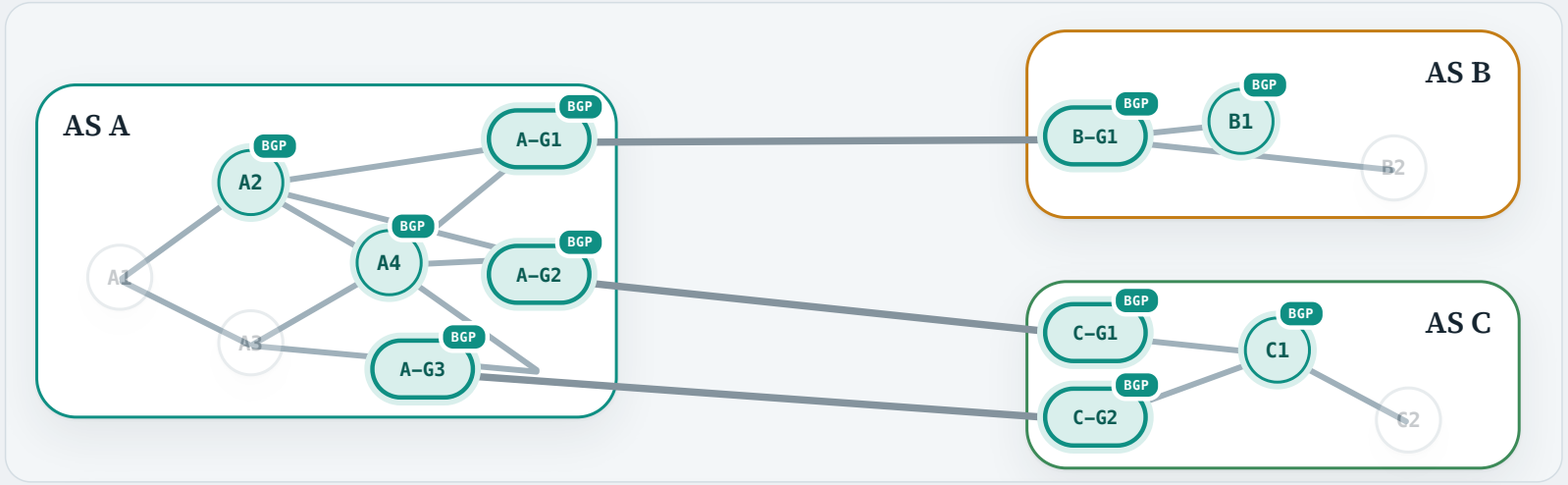
NOT ONLY GATEWAYS

BGP is not limited to gateways: internal routers such as A2, A4, B1, and C1 can also be BGP speakers.

BGP speaker definition: RFC 4271 §1.1

WHO PARTICIPATES IN BGP?

BGP is run by BGP-speaking routers

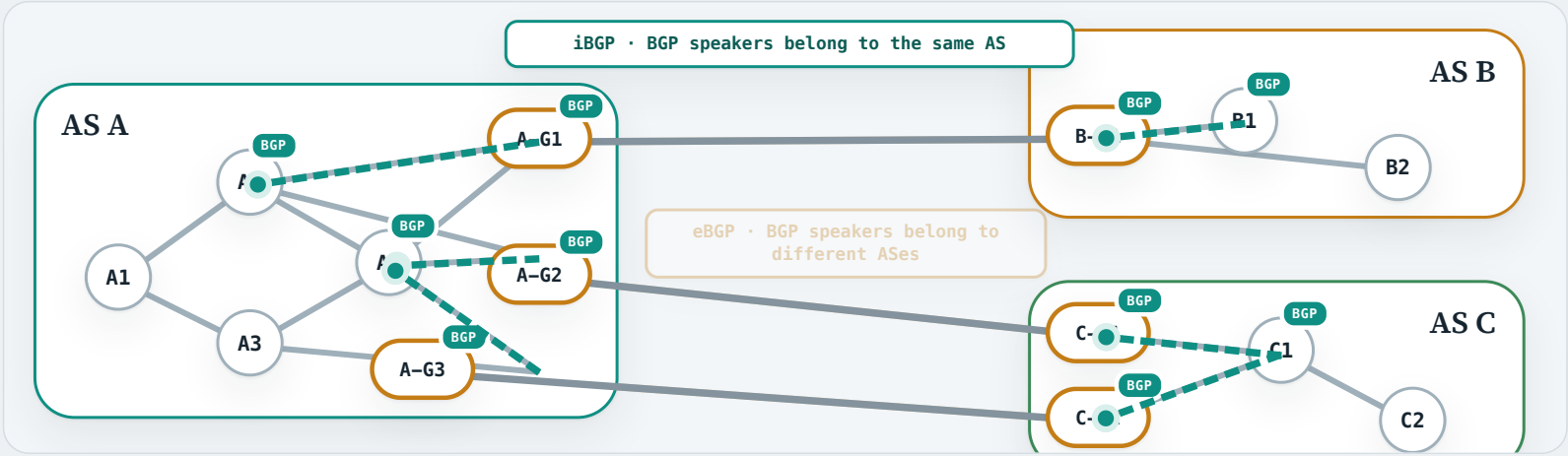


NOT EVERY NODE Only selected routers run BGP; ordinary hosts and many internal routers do not participate.

BGP speaker definition: RFC 4271 §1.1

TWO PEER RELATIONSHIPS · THE SAME PROTOCOL

A BGP session is external or internal



External BGP (eBGP)

this is how an AS **learns external reachability**

Sessions between BGP speakers in **different ASes**, usually the border speakers each side of the boundary. Using **eBGP**, an AS learns **AS-level paths** to destinations outside itself.

Internal BGP (iBGP)

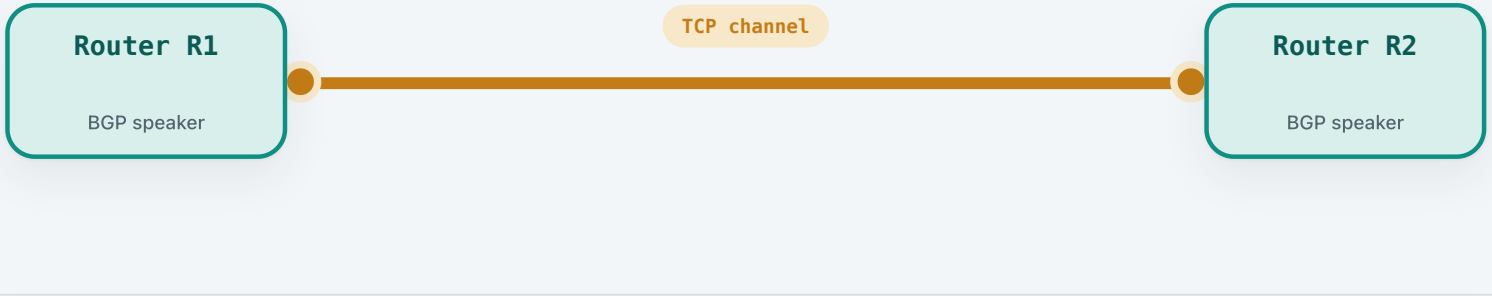
this is how the AS **distributes it inside**

Sessions between BGP speakers in the **same AS**. Using **iBGP**, all the routers inside the AS come to know those external paths, so each of them can forward towards the outside.

eBGP, iBGP and BGP speaker definitions: RFC 4271 §§1.1, 3

A STRUCTURED CONVERSATION BETWEEN TWO BGP SPEAKERS

A BGP session is a structured conversation



TCP · THE CHANNEL

Both iBGP and eBGP use TCP as the channel for the conversation. We will study TCP later in this course.

BGP transport and messages: RFC 4271 §§4.1–4.4, 8

A STRUCTURED CONVERSATION BETWEEN TWO BGP SPEAKERS

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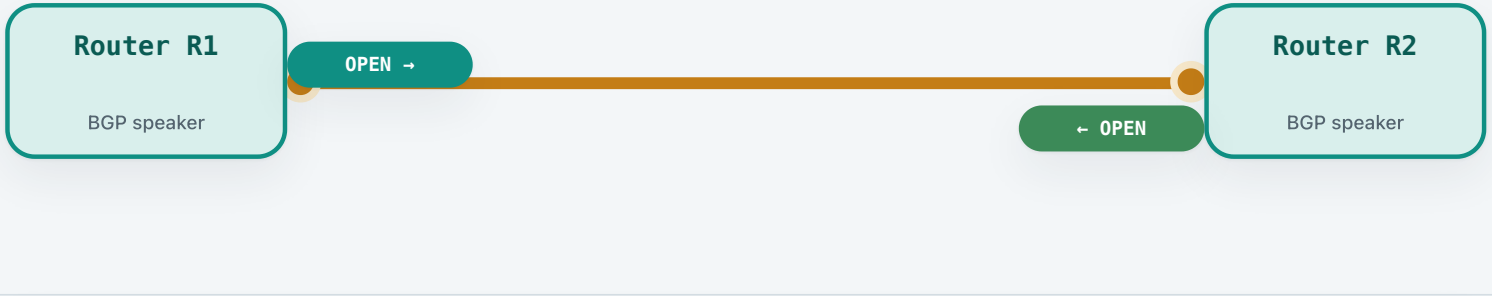


OPEN · ESTABLISH After TCP connects, OPEN messages establish the BGP relationship.

BGP transport and messages: RFC 4271 §§4.1–4.4, 8

A STRUCTURED CONVERSATION BETWEEN TWO BGP SPEAKERS

A BGP session is a structured conversation

KEEPALIVE · MAINTAIN

KEEPALIVE acknowledges OPEN and later confirms that the session remains active.

BGP transport and messages: RFC 4271 §§4.1–4.4, 8

A STRUCTURED CONVERSATION BETWEEN TWO BGP SPEAKERS

A BGP session is a structured conversation

UPDATE · EXCHANGE ROUTES

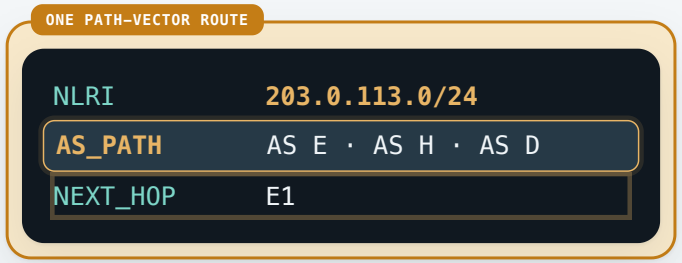
An UPDATE **advertises** one or more destination prefixes together with the **AS-level path** to them — that pairing is the **path vector**, which we open up next. An UPDATE can also **withdraw** prefixes that are no longer reachable.

BGP transport and messages: RFC 4271 §§4.1–4.4, 8

THE ROUTE INFORMATION CARRIED BY AN UPDATE

A closer look at a BGP path vector

Unlike a distance-vector protocol, what BGP advertises to a neighbour is a path vector.

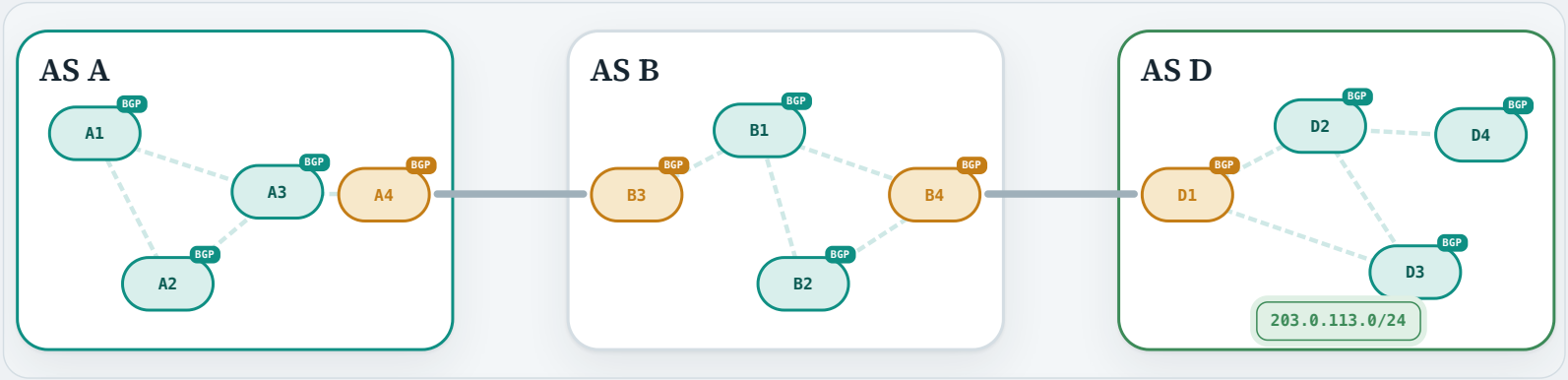


NLRI	**Network Layer Reachability Information:** the destination prefix being advertised.
AS_PATH	The complete **AS-level path leading to that destination prefix**.
NEXT_HOP	The IP address of the next router to which traffic should be forwarded towards that prefix.

BGP route, NLRI and path attributes: RFC 4271 §§1.1, 3.1, 4.3, 5.1.2–5.1.3

WORKED EXAMPLE · ONE ROUTE FROM AS D TO AS A

Watch one BGP route travel across three ASes



1 · AS D OWNS THE PREFIX

`203.0.113.0/24` is one of the subnets inside AS D. For hosts elsewhere on the Internet to reach it, AS D has to **advertise this destination prefix to the other ASes**.

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

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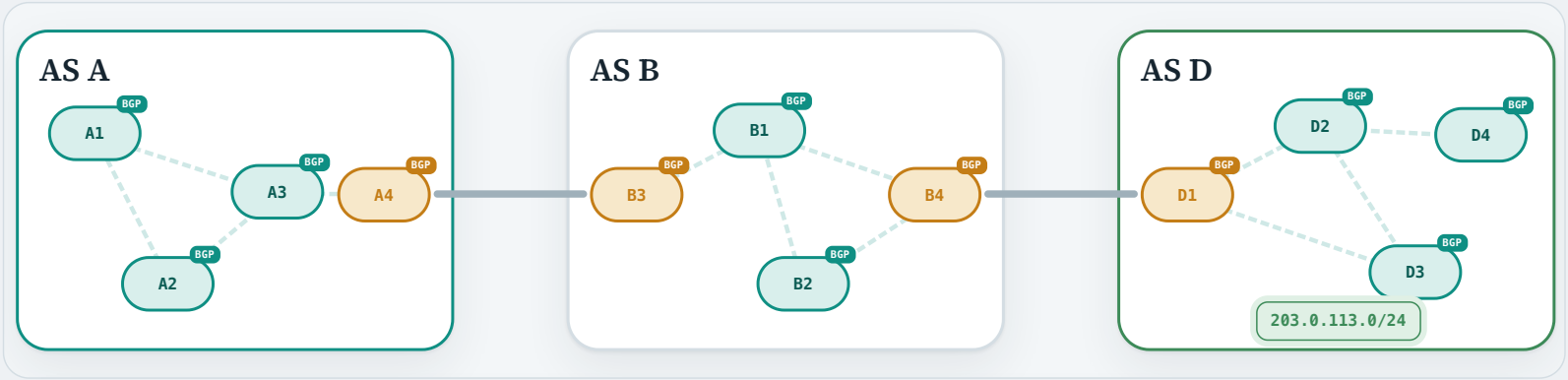

2 · AS D EXPORT POLICY

AS D has a neighbour it could advertise to — AS B. But before it does, it consults its **export policy**: is advertising this prefix to this neighbour something AS D is willing to do?

We will return to export policy later.

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

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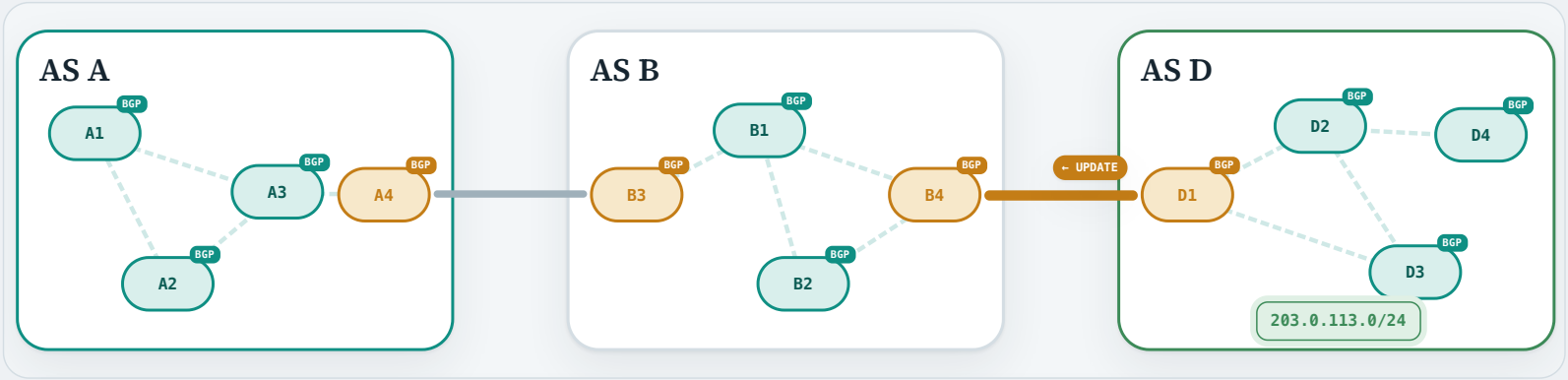

3 · EBGP: D → B

Suppose AS D permits the export. D1 sends an UPDATE to B4 over their established eBGP session.

```
NLRI        203.0.113.0/24
AS_PATH     AS D
NEXT_HOP    D1
```

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

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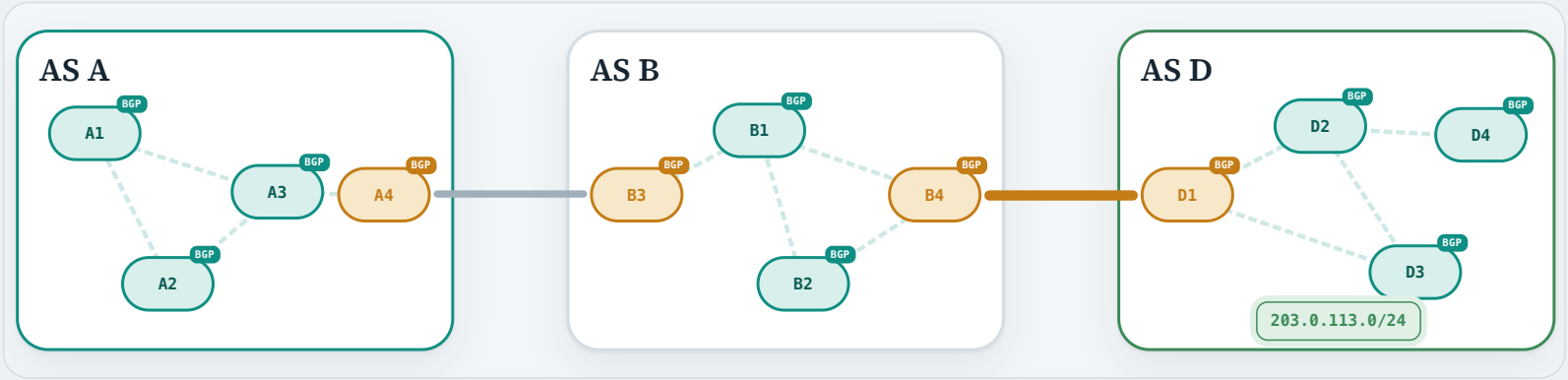

4 · AS B IMPORT POLICY

AS B applies its **import policy**. B4 may accept or reject D1's advertisement.

We will return to import policy later.

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

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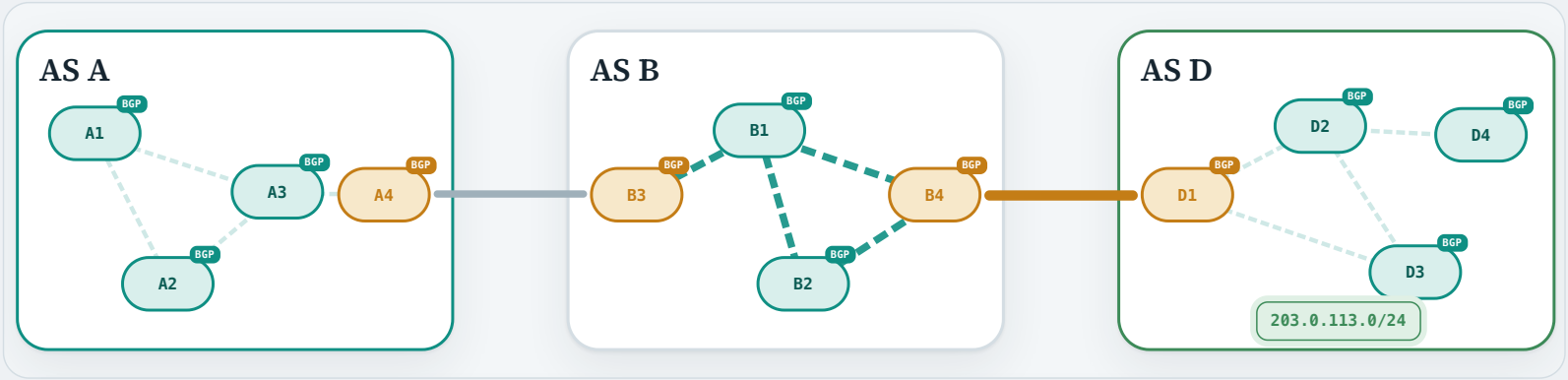


5 · IBGP INSIDE B

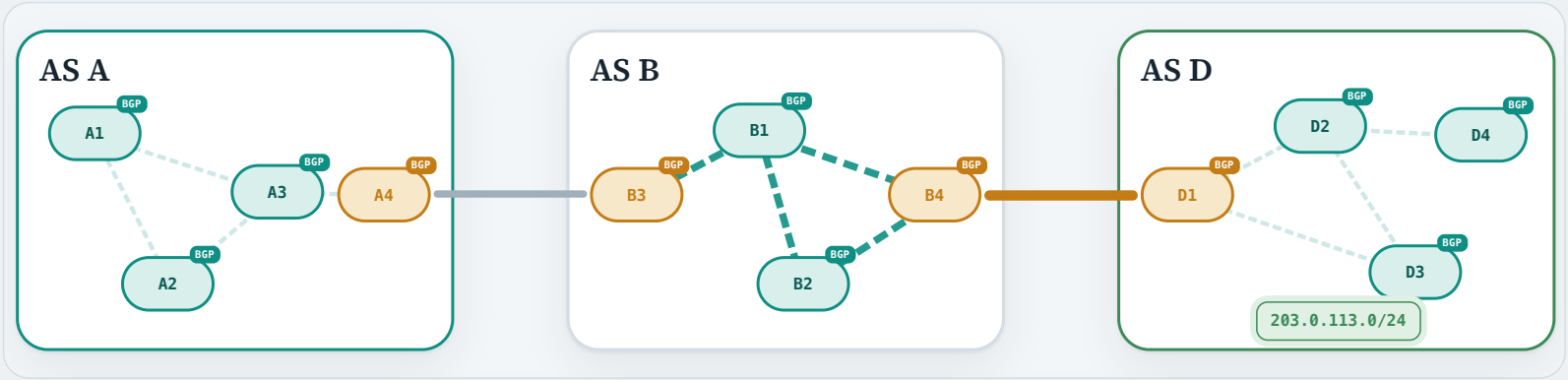
Suppose AS B accepts. B4 then shares the accepted route using **internal BGP (iBGP)** with AS B's other BGP speakers; in particular, B3 learns it.

```
NLRI        203.0.113.0/24
AS_PATH     AS D · unchanged
NEXT_HOP    D1
```

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

WORKED EXAMPLE · ONE ROUTE FROM AS D TO AS A

Watch one BGP route travel across three ASes



6 · AS B EXPORT POLICY

AS B now knows it can reach the prefix, and it would like its own neighbours to know that too. It has a neighbour it could advertise to — AS A. But first it consults its **export policy**: is carrying other ASes' traffic to this prefix something AS B is willing to do?

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

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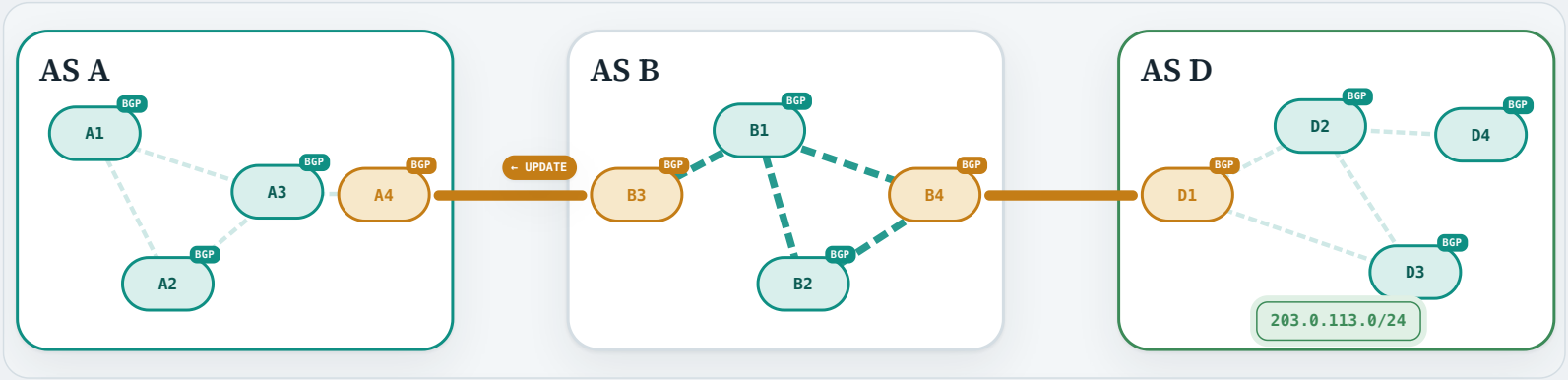

7 · EBGP: B → A

Suppose AS B permits the export. B3 **prepends AS B** onto the `AS_PATH` and sets `NEXT_HOP` to its own IP address, then sends this path vector in an UPDATE to AS A's border router A4.

```
NLRI        203.0.113.0/24
AS_PATH     AS B · AS D
NEXT_HOP    B3
```

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

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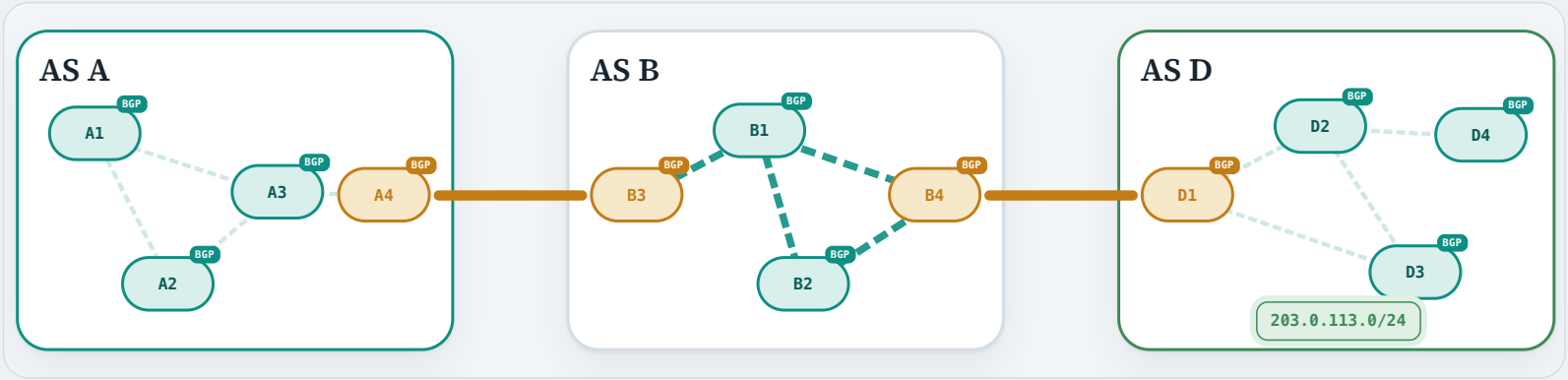

8 · AS A IMPORT POLICY

AS A applies its **import policy**. A4 may accept or reject B3's advertisement.

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

WORKED EXAMPLE · ONE ROUTE FROM AS D TO AS A

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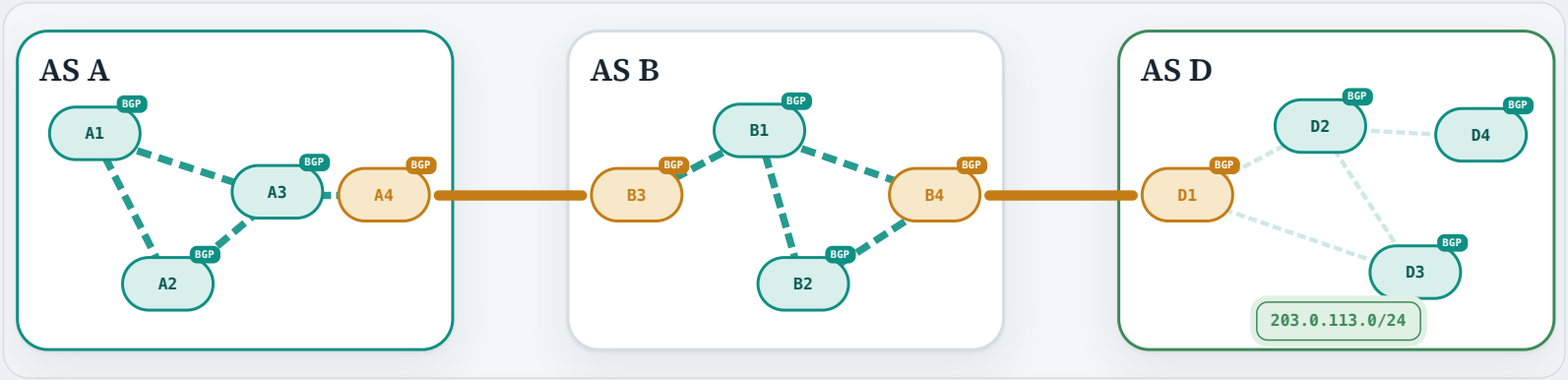

9 · IBGP INSIDE A

Suppose AS A accepts. A4 then shares the accepted route using **internal BGP (iBGP)** with AS A's other BGP speakers.

```
NLRI        203.0.113.0/24
AS_PATH     AS B · AS D
NEXT_HOP    B3
```

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

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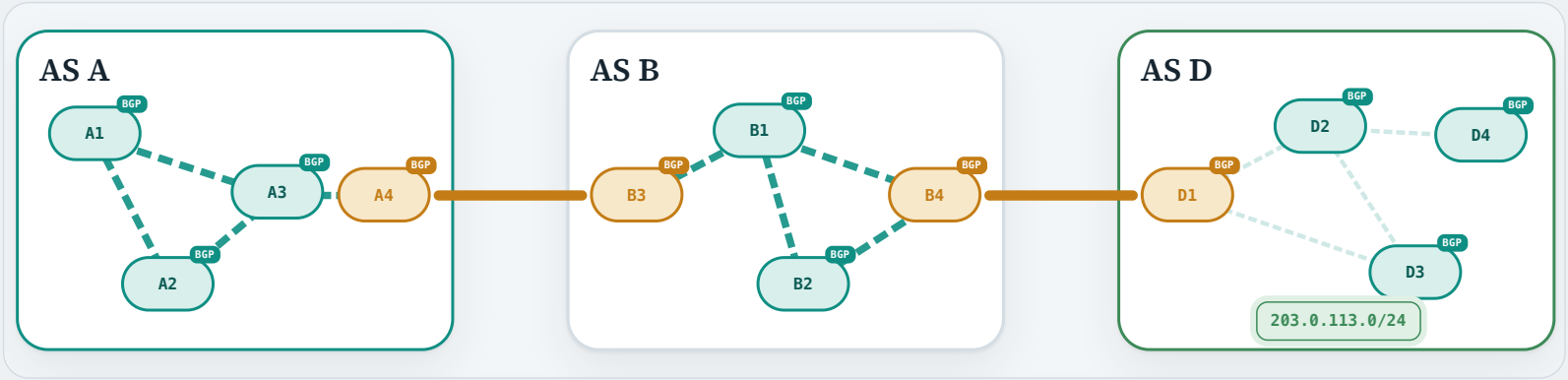


10 · EVERY ROUTER IN AS A KNOWS THE WAY

Every BGP speaker in AS A now knows that `203.0.113.0/24` is reachable **via AS B, then AS D** — carried between the ASes by **eBGP** and spread inside each one by **iBGP**.

BGP UPDATE, path attributes, and import/export policy: RFC 4271 §§3, 4.3, 5.1.2–5.1.3, 9.1–9.2

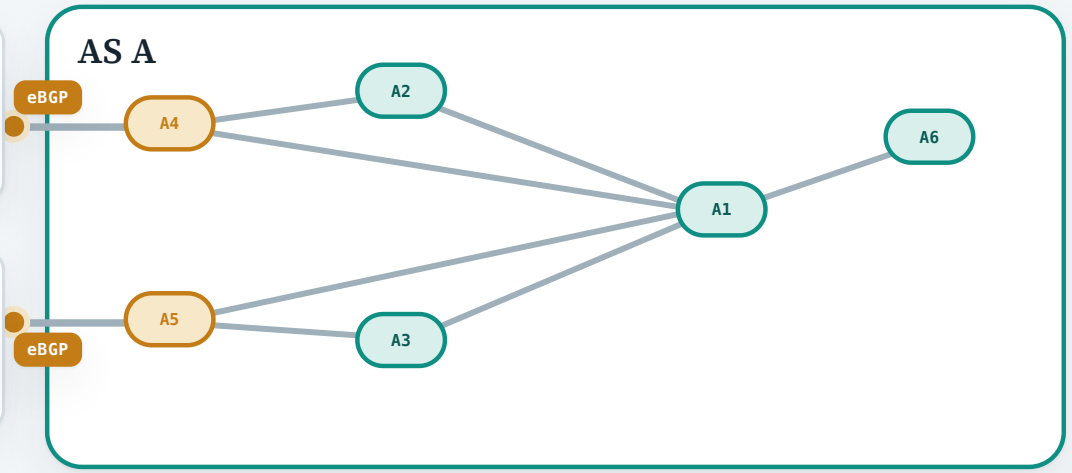
eBGP import policy: which routes an AS prefers

ADVERTISEMENT FROM AS B

NLRI	203.0.113.0/24
AS_PATH	AS B · AS D
NEXT_HOP	B3

ADVERTISEMENT FROM AS E

NLRI	203.0.113.0/24
AS_PATH	AS E · AS H · AS D
NEXT_HOP	E1

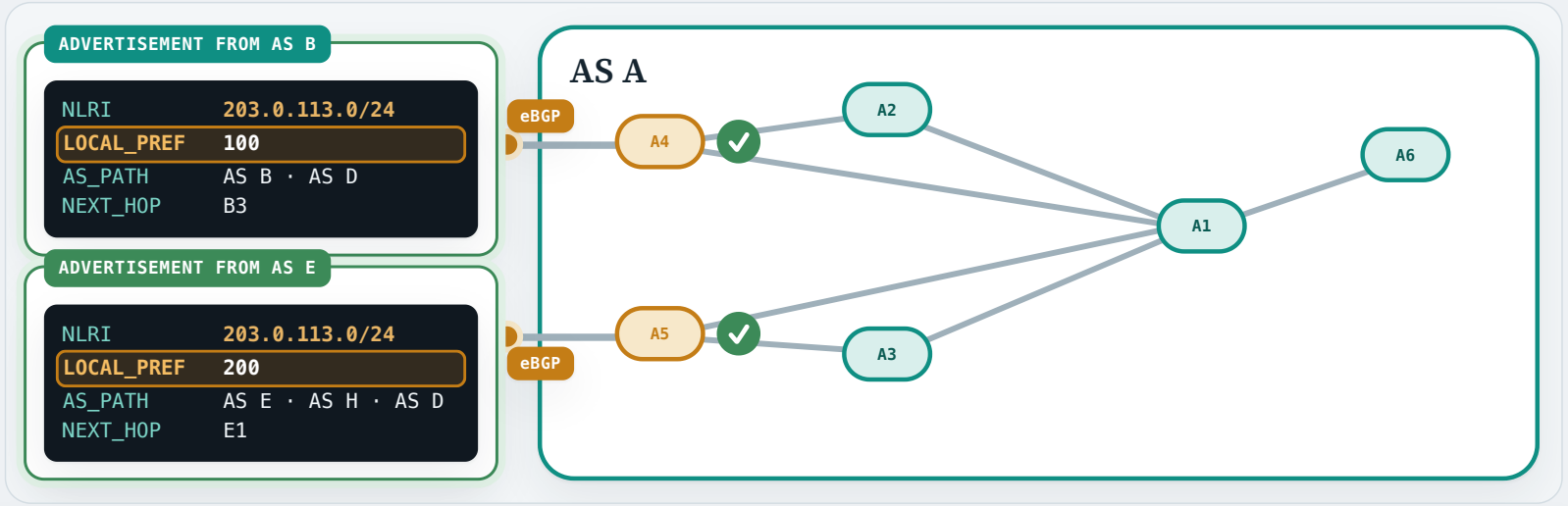


1 · ONE PREFIX, TWO ROUTES

Two advertisements for the same prefix, `203.0.113.0/24`, arrive over separate eBGP sessions at two different AS A border gateways. **Which one should AS A allow to import?**

Import policy and LOCAL_PREF: RFC 4271 §§5.1.5, 9.1.1

eBGP import policy: which routes an AS prefers

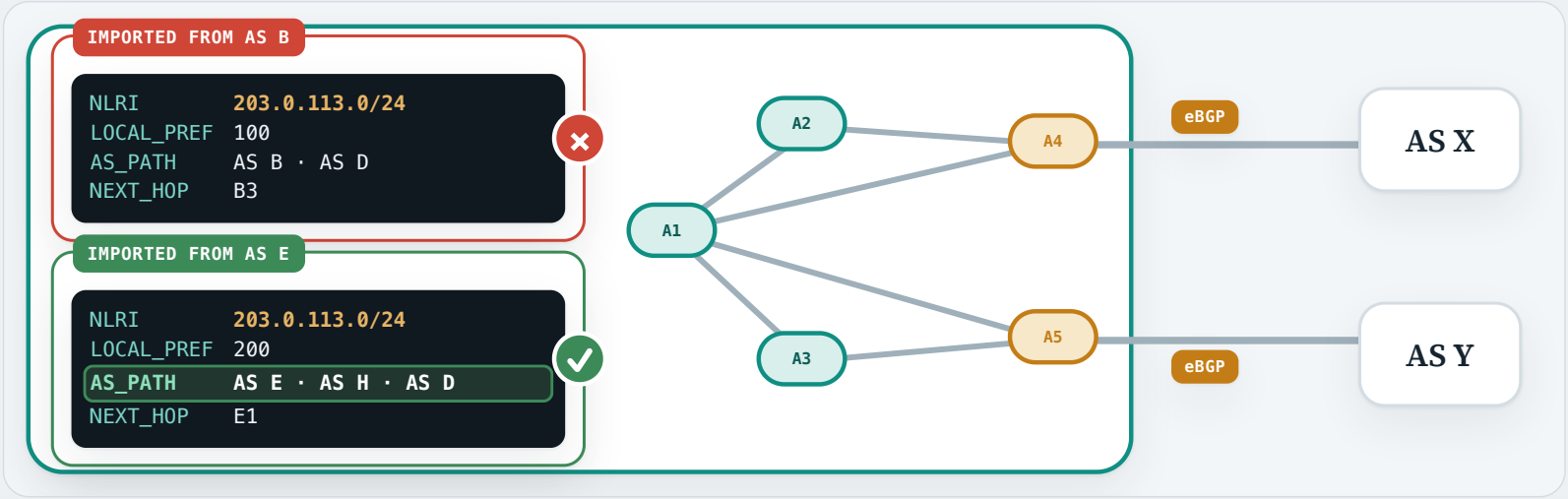


2 · ACCEPT BOTH, AND RANK THEM

AS A's import policy accepts both routes and attaches a `LOCAL_PREF` to each — 100 and 200 — for use inside AS A. The higher value is the one AS A prefers.

Import policy and LOCAL_PREF: RFC 4271 §§5.1.5, 9.1.1

eBGP export policy: what an AS advertises



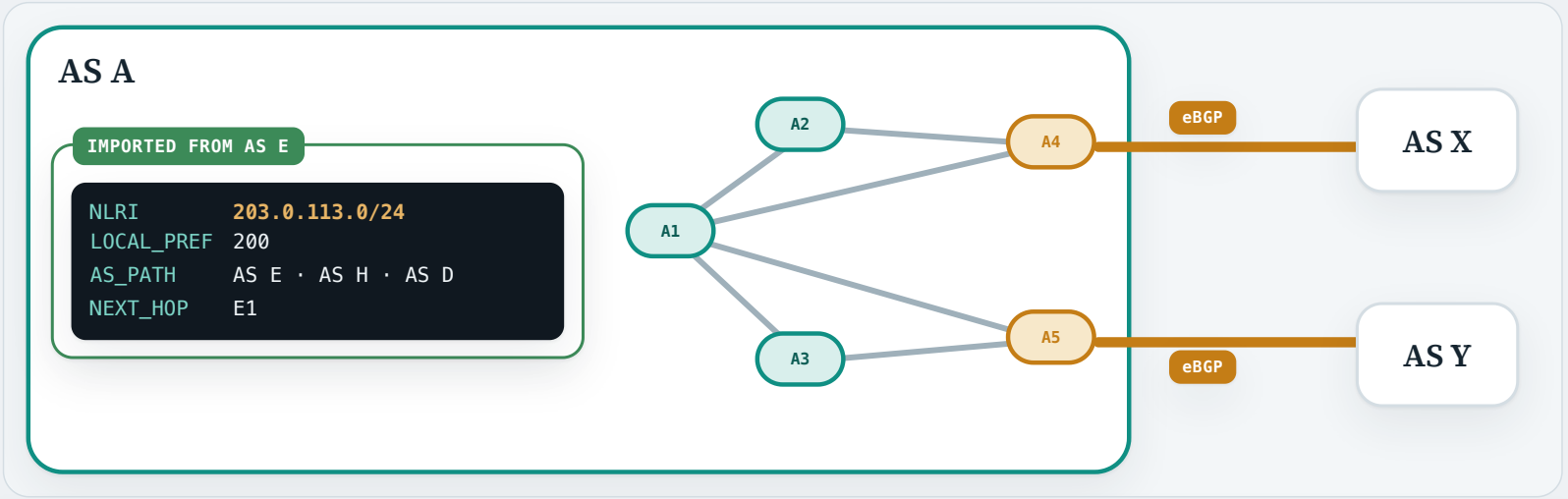
1 · SELECT AN AS PATH

Both routes were imported, but the one through AS E carries the higher `LOCAL_PREF`, so AS A selects **AS E · AS H · AS D** for `203.0.113.0/24`.

Per-neighbour export policy and route dissemination: RFC 4271 §§9.2, 5.1.5; RFC 8212

AS-WIDE ROUTING INTENTION · OUTGOING ADVERTISEMENTS

eBGP export policy: what an AS advertises



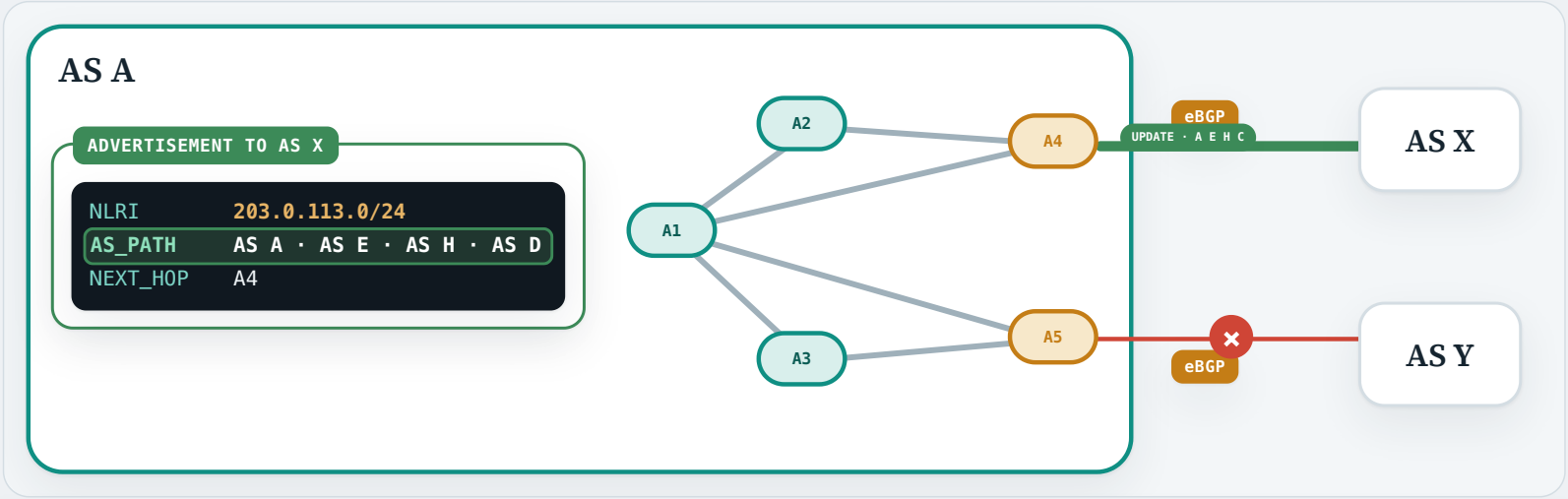
2 · ADVERTISE IT ONWARD?

AS A can now reach the prefix, and it has two neighbours it could tell — AS X and AS Y. **Which of them should AS A advertise this route to?**

Per-neighbour export policy and route dissemination: RFC 4271 §§9.2, 5.1.5; RFC 8212

AS-WIDE ROUTING INTENTION · OUTGOING ADVERTISEMENTS

eBGP export policy: what an AS advertises



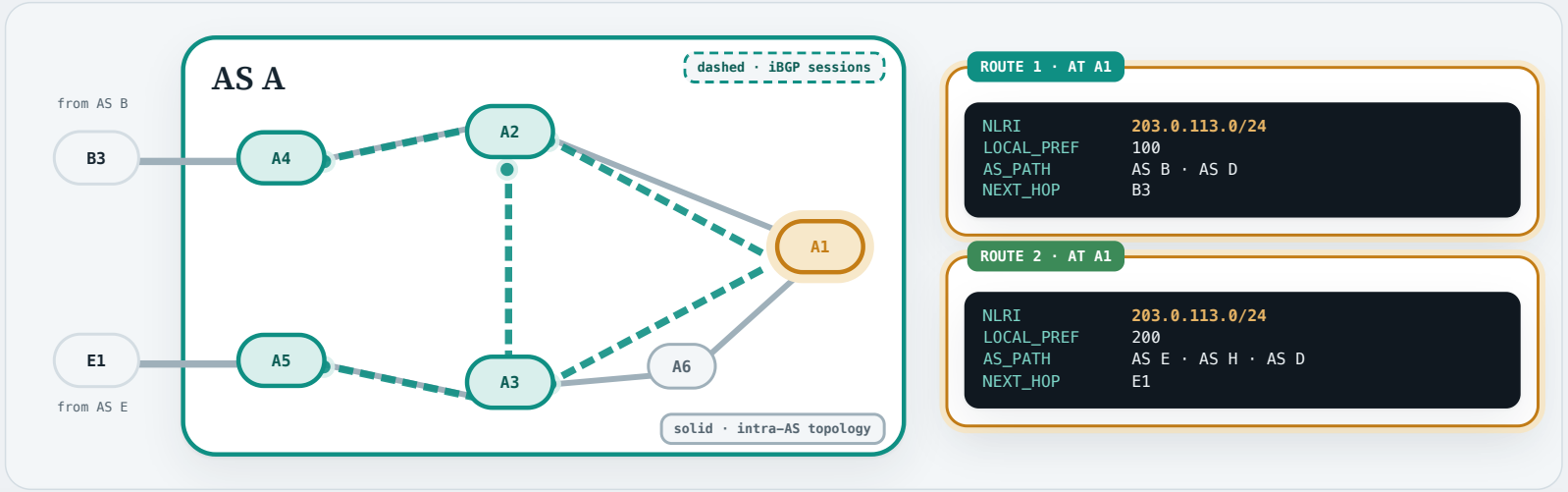
3 · EXPORT POLICY DECIDES

AS A's export policy answers per neighbour: it prepends itself and advertises **AS A · AS E · AS H · AS D** to AS X, while withholding it from AS Y.

Per-neighbour export policy and route dissemination: RFC 4271 §§9.2, 5.1.5; RFC 8212

ONE NEXT HOP PER PREFIX · LOCAL_PREF → AS_PATH → INTRA-AS COST

How each BGP router builds its forwarding table



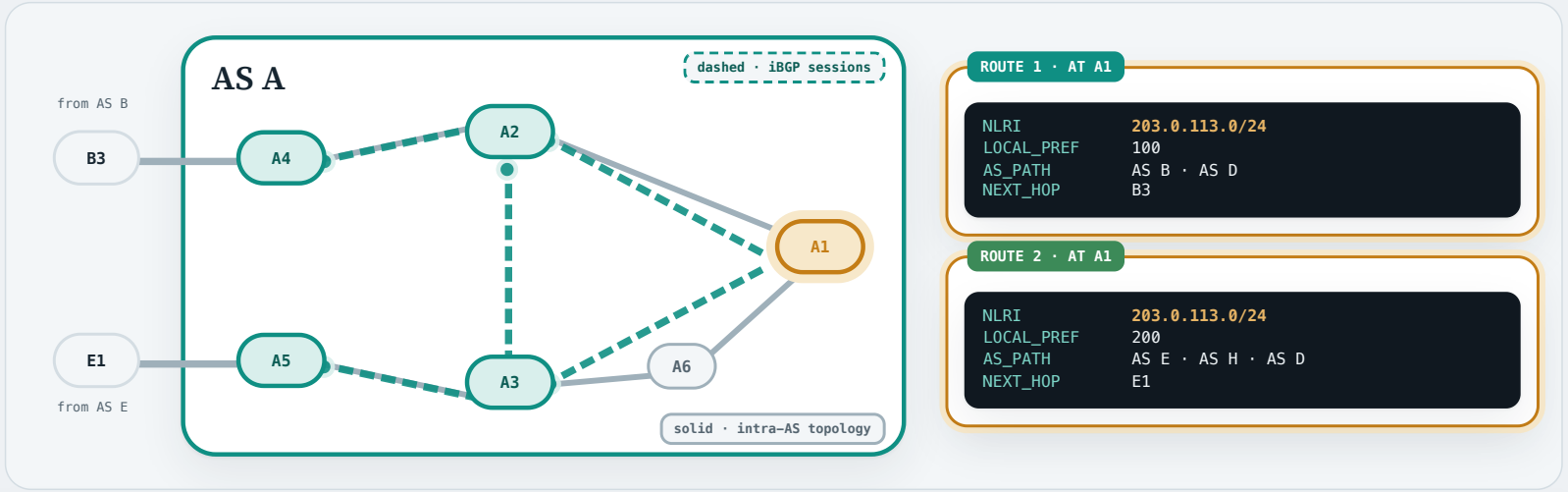
THE SETUP

Over iBGP, router A1 has learned **two routes** to `203.0.113.0/24` — one from A4, one from A5.

iBGP, distributed selection, LOCAL_PREF, and IGP cost to NEXT_HOP: RFC 4271 §§3, 5.1.5, 9.1.1–9.1.2

ONE NEXT HOP PER PREFIX · LOCAL_PREF → AS_PATH → INTRA-AS COST

How each BGP router builds its forwarding table



THE QUESTION

A1 must put **one** next hop for this prefix into its forwarding table. **Which of the two routes should it pick?**

iBGP, distributed selection, LOCAL_PREF, and IGP cost to NEXT_HOP: RFC 4271 §§3, 5.1.5, 9.1.1–9.1.2

ONE NEXT HOP PER PREFIX · LOCAL_PREF → AS_PATH → INTRA-AS COST

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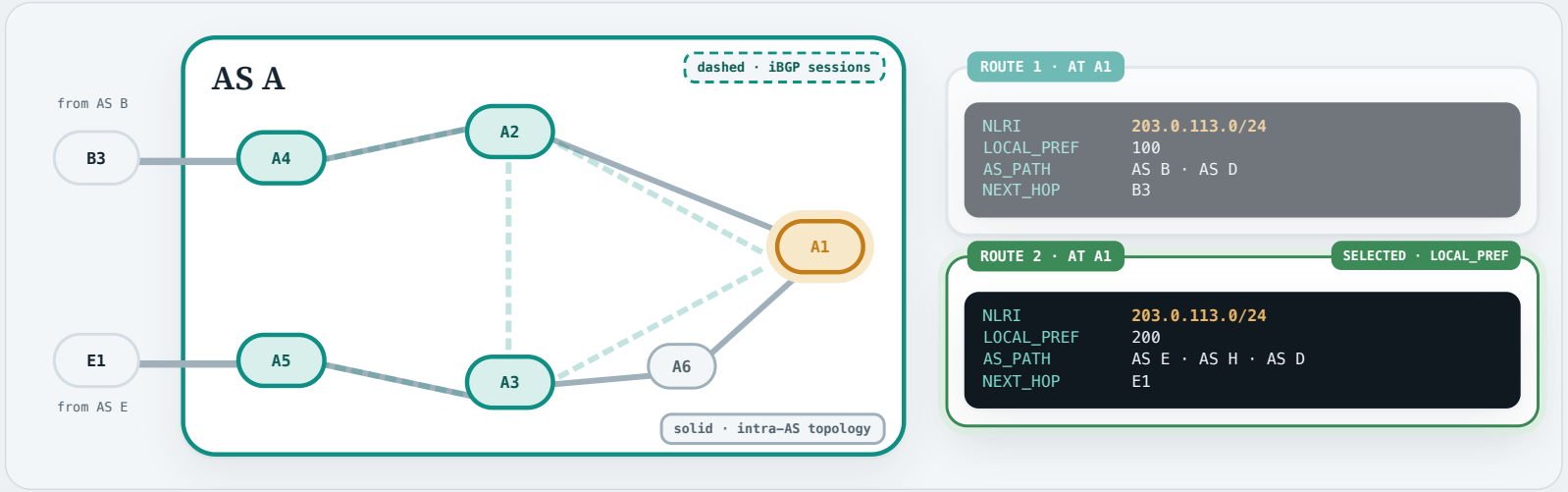
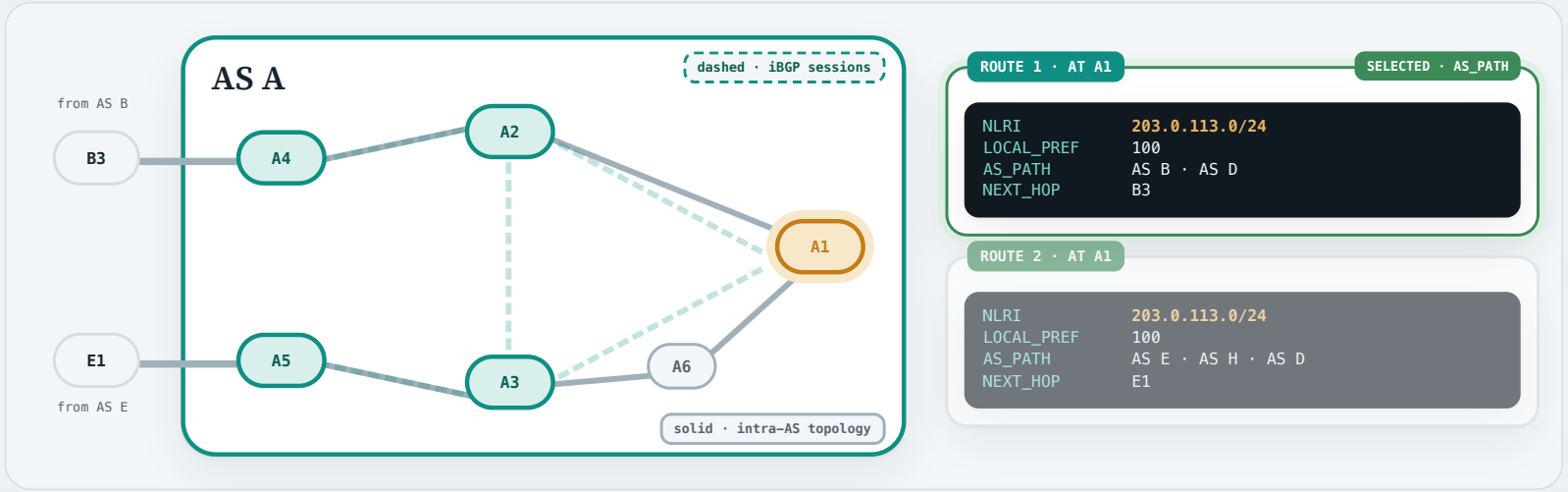


1 · AS-WIDE POLICY FIRST

The first tie-breaker is `LOCAL_PREF`, the AS-wide preference from import policy: 200 beats 100, so **Route 2 wins**.

iBGP, distributed selection, LOCAL_PREF, and IGP cost to NEXT_HOP: RFC 4271 §§3, 5.1.5, 9.1.1–9.1.2

How each BGP router builds its forwarding table

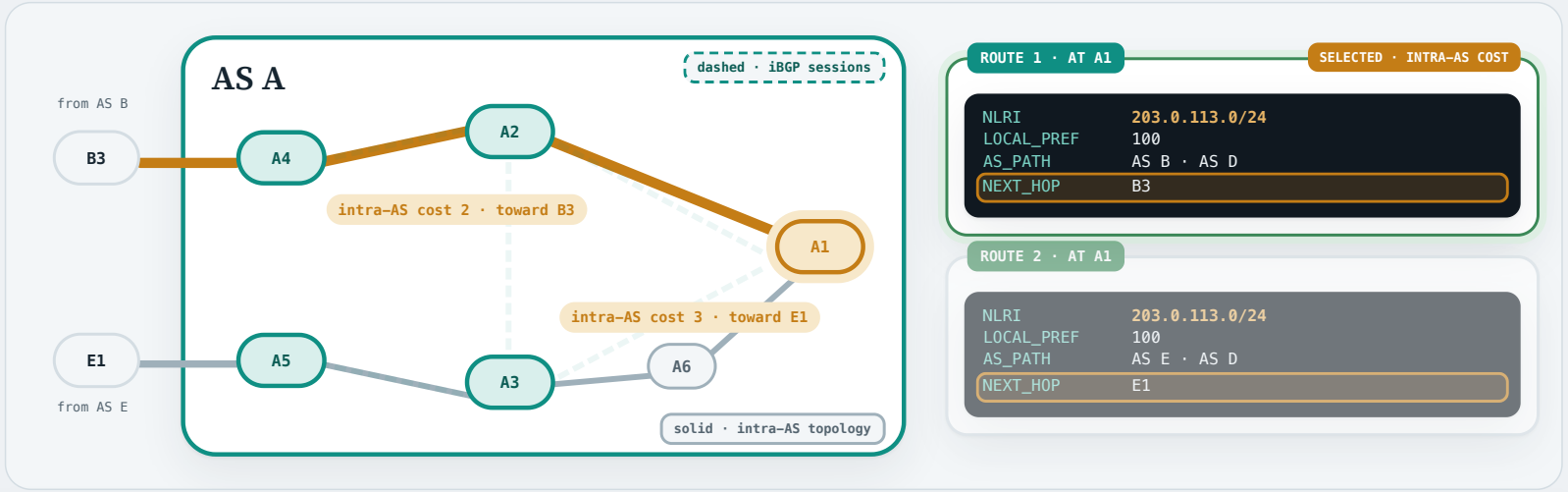


2 · THEN AS_PATH LENGTH

If `LOCAL_PREF` ties, A1 compares AS-level path lengths and takes the shorter: **Route 1 wins** with two ASes rather than three.

iBGP, distributed selection, LOCAL_PREF, and IGP cost to NEXT_HOP: RFC 4271 §§3, 5.1.5, 9.1.1–9.1.2

How each BGP router builds its forwarding table



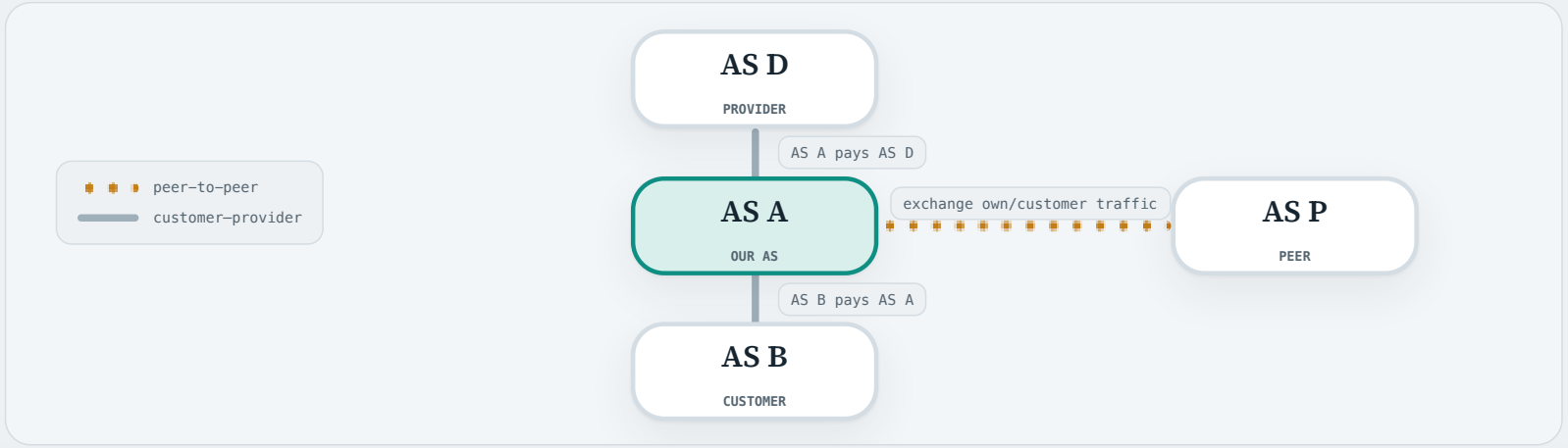
3 · THEN HOT-POTATO ROUTING

If both BGP attributes tie, A1 turns to AS A's **own internal topology** and asks its own intra-AS routing protocol which next hop is closer: B3 at cost 2 beats E1 at cost 3, so **Route 1 wins — hot-potato routing**.

iBGP, distributed selection, LOCAL_PREF, and IGP cost to NEXT_HOP: RFC 4271 §§3, 5.1.5, 9.1.1–9.1.2

THE BUSINESS MODEL BEHIND INTER-AS ROUTE POLICY

Business relationships between ASes



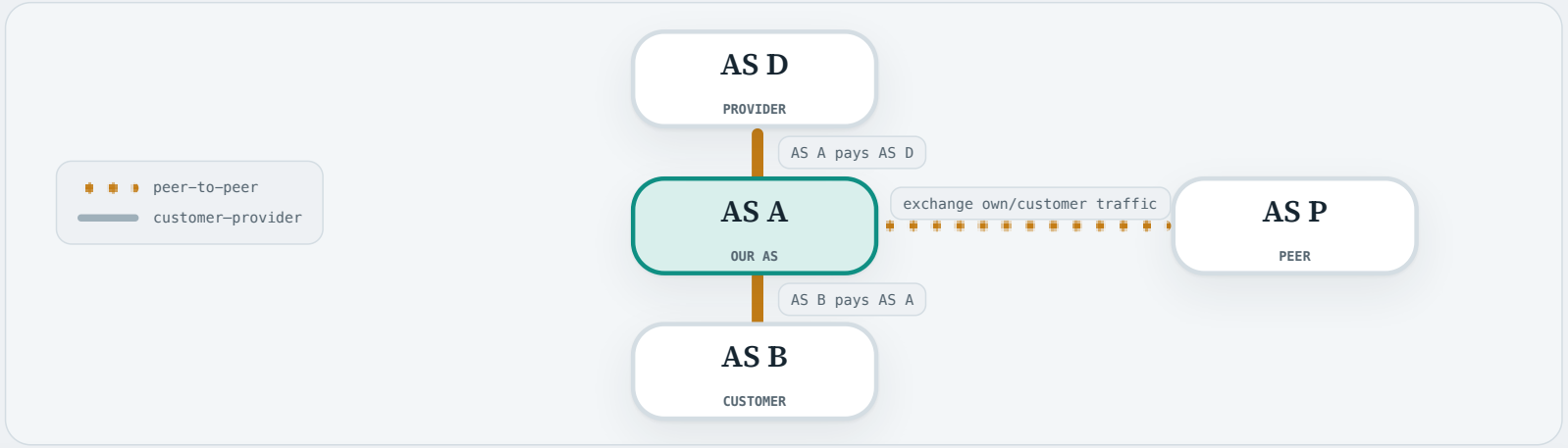
THE QUESTION

What drives an AS's import and export policy?
The answer lies in the **business relationships** between ASes — who pays whom for carrying traffic.

Customer–provider and peer relationships: CAIDA AS Relationships; RFC 9234

THE BUSINESS MODEL BEHIND INTER-AS ROUTE POLICY

Business relationships between ASes



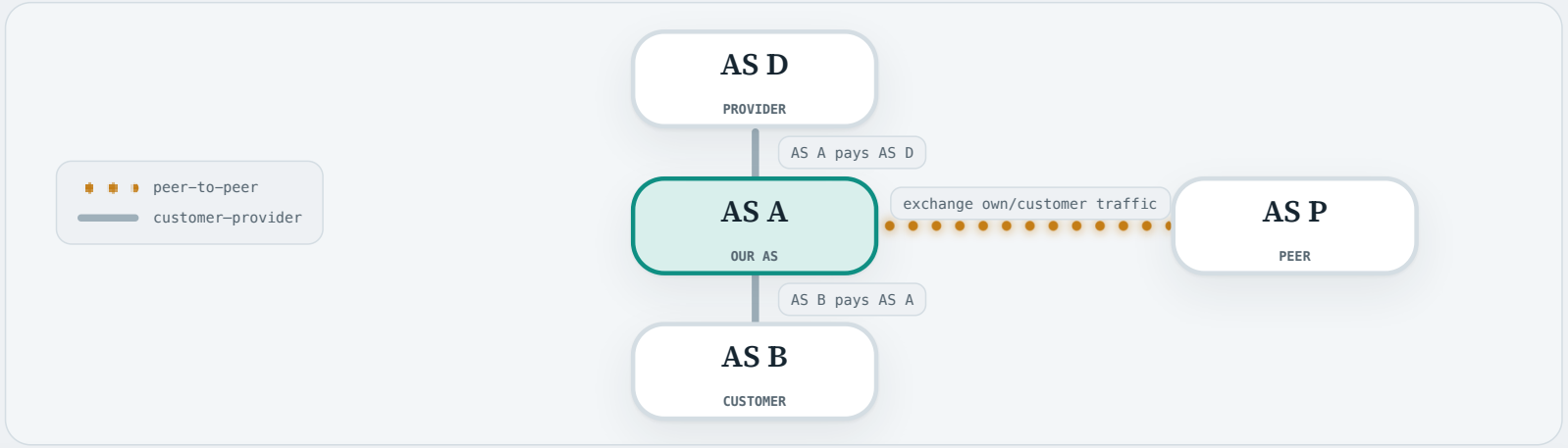
CUSTOMER → PROVIDER

The customer pays the provider for transit to the wider Internet.
e.g. AS A buys its connectivity to the rest of the Internet from AS D, and pays AS D for it.

Customer–provider and peer relationships: CAIDA AS Relationships; RFC 9234

THE BUSINESS MODEL BEHIND INTER-AS ROUTE POLICY

Business relationships between ASes

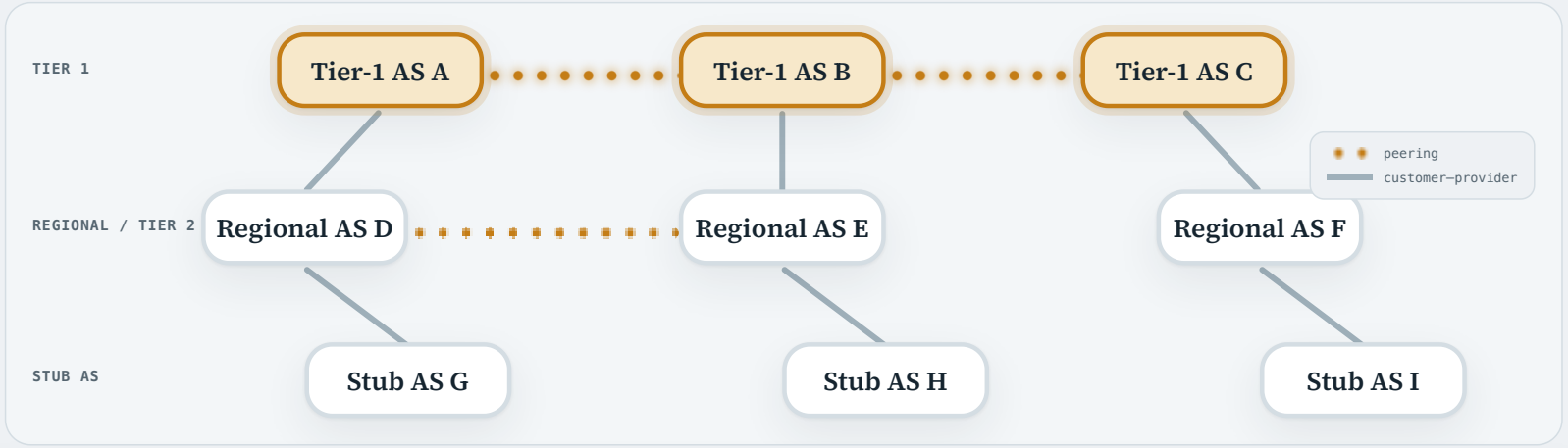


PEER ↔ PEER	Peers commonly exchange traffic for themselves and their customers—not general transit for one another. *e.g.* AS A and AS P exchange roughly as much traffic as each other, so neither pays the other and neither depends on the other for connectivity.

Customer–provider and peer relationships: CAIDA AS Relationships; RFC 9234

HOW CUSTOMER, PROVIDER, AND PEER RELATIONSHIPS REPEAT ACROSS THE INTERNET

These relationships create a commercial AS hierarchy

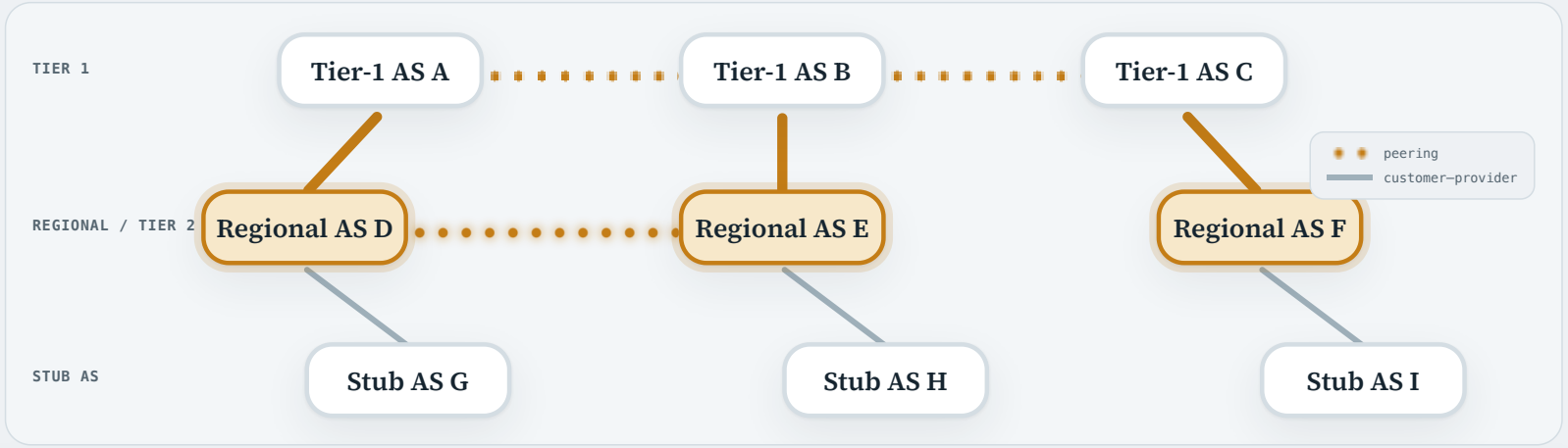


TIER-1 TRANSIT ASES They commonly peer with one another and do not purchase upstream transit.

Simplified AS hierarchy and valley-free path model: CAIDA AS Relationships; Gao, 2001

HOW CUSTOMER, PROVIDER, AND PEER RELATIONSHIPS REPEAT ACROSS THE INTERNET

These relationships create a commercial AS hierarchy

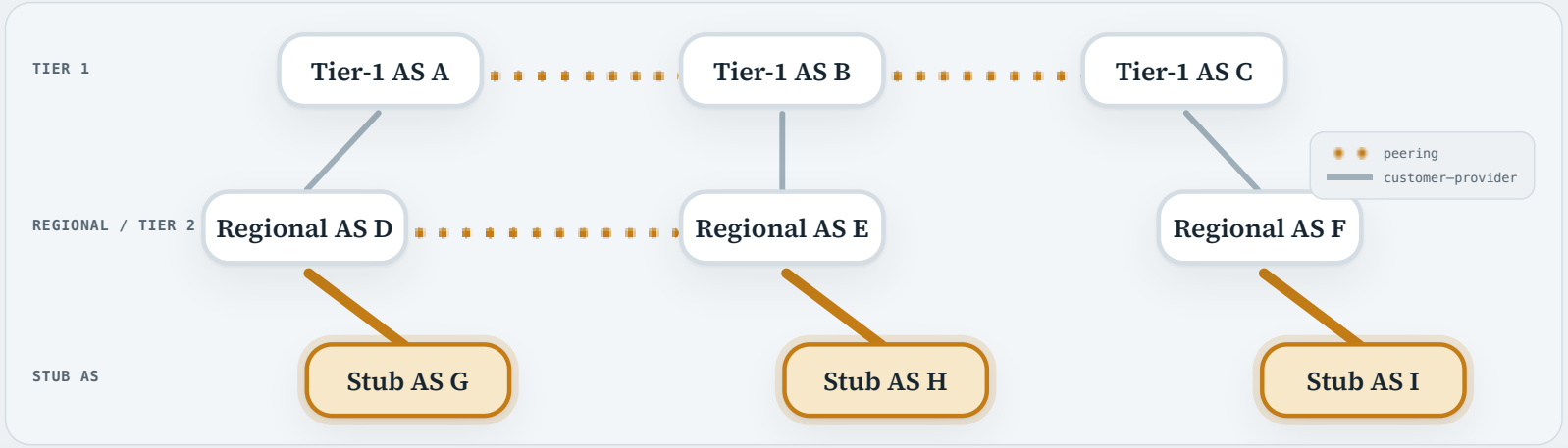


REGIONAL TRANSIT ASES They may buy transit upstream, sell it downstream, and peer with other regional ASes.

Simplified AS hierarchy and valley-free path model: CAIDA AS Relationships; Gao, 2001

HOW CUSTOMER, PROVIDER, AND PEER RELATIONSHIPS REPEAT ACROSS THE INTERNET

These relationships create a commercial AS hierarchy

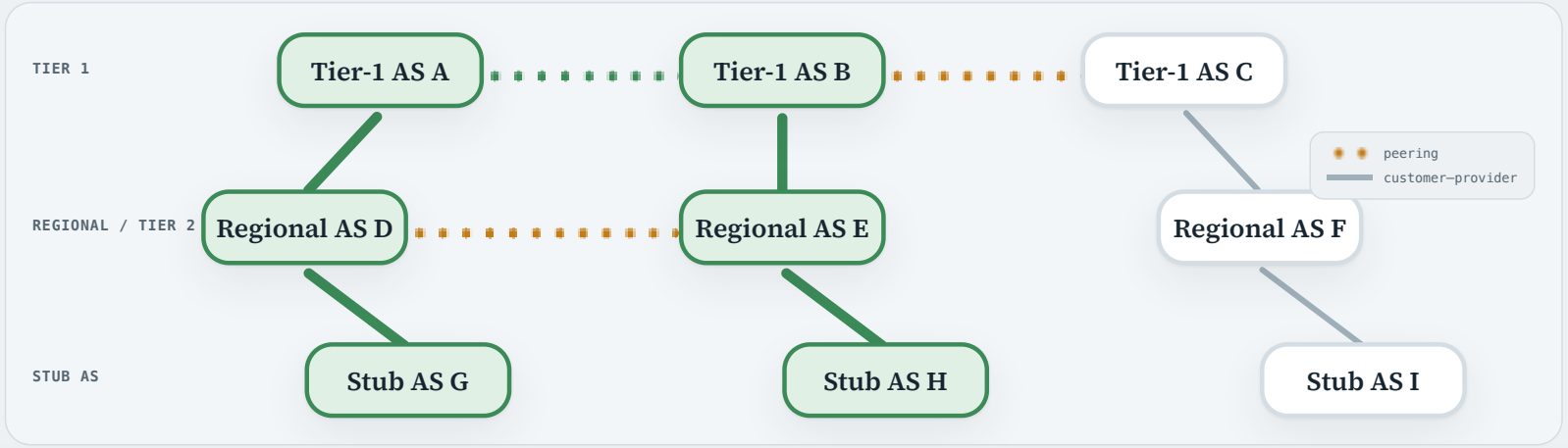


STUB ASES A stub reaches the wider Internet through providers and does not carry transit between other ASes.

Simplified AS hierarchy and valley-free path model: CAIDA AS Relationships; Gao, 2001

HOW CUSTOMER, PROVIDER, AND PEER RELATIONSHIPS REPEAT ACROSS THE INTERNET

These relationships create a commercial AS hierarchy



VALLEY–FREE PATHS

Whenever every AS's import and export policy is guided by these relationships, a path takes an **up–across–down** shape — a **valley-free path**.
e.g. a datagram from stub AS G to stub AS H climbs through providers to the Tier-1 level, crosses one peer link, then descends to AS H.

Simplified AS hierarchy and valley–free path model: CAIDA AS Relationships; Gao, 2001

A COMMON COMMERCIAL ORDERING—NOT FIXED BGP VALUES

Import policy: which route should the AS prefer?

ROUTE LEARNED FROM	COMMON PREFERENCE	ECONOMIC INTUITION
Customer	HIGH	The customer pays us for transit.
Peer	MEDIUM	Usually settlement-free exchange.
Provider	LOW	We pay the provider for transit.

1 · CUSTOMER ROUTE Customer traffic earns revenue, so customer-learned routes commonly receive the highest preference.

LOCAL_PREF: RFC 4271 §5.1.5; common commercial ordering: Gao–Rexford policy model

A COMMON COMMERCIAL ORDERING—NOT FIXED BGP VALUES

Import policy: which route should the AS prefer?

ROUTE LEARNED FROM	COMMON PREFERENCE	ECONOMIC INTUITION
Customer	HIGH	The customer pays us for transit.
Peer	MEDIUM	Usually settlement-free exchange.
Provider	LOW	We pay the provider for transit.

2 · PEER ROUTE A peer route is commonly preferred over a route that incurs upstream transit cost.

LOCAL_PREF: RFC 4271 §5.1.5; common commercial ordering: Gao–Rexford policy model

A COMMON COMMERCIAL ORDERING—NOT FIXED BGP VALUES

Import policy: which route should the AS prefer?

ROUTE LEARNED FROM	COMMON PREFERENCE	ECONOMIC INTUITION
Customer	HIGH	The customer pays us for transit.
Peer	MEDIUM	Usually settlement-free exchange.
Provider	LOW	We pay the provider for transit.

3 · PROVIDER ROUTE

Provider-learned routes commonly receive low preference because the AS must pay the provider to carry that traffic.

LOCAL_PREF: RFC 4271 §5.1.5; common commercial ordering: Gao–Rexford policy model

A COMMON COMMERCIAL GUIDELINE—NOT A RULE ENFORCED BY BGP

Export policy: where should the route be advertised?

ROUTE LEARNED FROM	TO CUSTOMER	TO PEER	TO PROVIDER
Customer	✓	✓	✓
Peer	✓	✗	✗
Provider	✓	✗	✗

TO CUSTOMERS Customers pay for transit, so they can commonly receive every usable route.

Common relationship-based propagation guideline: RFC 7908; relationship roles: RFC 9234

A COMMON COMMERCIAL GUIDELINE—NOT A RULE ENFORCED BY BGP

Export policy: where should the route be advertised?

ROUTE LEARNED FROM	TO CUSTOMER	TO PEER	TO PROVIDER
Customer	✓	✓	✓
Peer	✓	×	×
Provider	✓	×	×

CUSTOMER-LEARNED ROUTES These can commonly be advertised to peers and providers because the customer pays the AS for transit.

Common relationship-based propagation guideline: RFC 7908; relationship roles: RFC 9234

A COMMON COMMERCIAL GUIDELINE—NOT A RULE ENFORCED BY BGP

Export policy: where should the route be advertised?

ROUTE LEARNED FROM	TO CUSTOMER	TO PEER	TO PROVIDER
Customer	✓	✓	✓
Peer	✓	×	×
Provider	✓	×	×

PEER/PROVIDER ROUTES Do not normally export these to another peer or provider; doing so creates unpaid transit.

Common relationship-based propagation guideline: RFC 7908; relationship roles: RFC 9234

Inter-AS routing recap

- **BGP solves the inter-AS routing problem** by learning AS-level paths to destination network prefixes.
- **eBGP exchanges routes between ASes**, while **iBGP distributes candidate routes inside an AS**.
- **Both use BGP's path-vector mechanism**, carrying a destination prefix together with its AS-level path and next hop.
- **Import and export policies express AS-wide routing choices**, often driven by customer, provider, and peer relationships.

MODULE 2 · ROUTING INSIDE ONE ADMINISTRATIVE DOMAIN

Intra-AS Routing Protocols

1 · Inter-AS routing and AS policy | **2 · Intra-AS routing protocols** | 3 · Aggregation, NAT and IPv6

ROUTING WITHIN ONE AUTONOMOUS SYSTEM

An overview of intra-AS routing protocols

RIP

Routing Information Protocol

DISTANCE VECTOR

OSPF

Open Shortest Path First

LINK STATE

IS-IS

Intermediate System to Intermediate System

LINK STATE

EIGRP

Enhanced Interior Gateway Routing Protocol

DISTANCE VECTOR

IGP definition: RFC 1812 §7.2.1; RIP, OSPF and IS-IS: RFC 6518; EIGRP: RFC 7868

A CONCRETE DISTANCE-VECTOR INTRA-AS PROTOCOL

RIP is a concrete distance-vector protocol

1 **A reminder: neighbour-to-neighbour exchange**

Routers talk only to their direct neighbours, and from that alone each learns the **best distance** and the **next hop** to every destination. No router ever learns the full topology.

2 **Every link costs one hop**

RIP measures distance in **router hops**: each router adds one to the metric it received.

distance from A to D = 3 hops

3 **15 hops is as far as RIP counts**

A path of more than **15 hops** is treated as unreachable — a metric of 16 means infinity.

RIP distance-vector operation and hop-count metric: RFC 2453 §§3.4, 3.9.2

RIP'S OWN DATABASE, AND THE TABLE THE DATAGRAMS USE

A RIP router keeps two tables

RIP routing table at A

destination	metric	next hop	timer
198.51.100.0/24	1	B	180 s
203.0.113.0/24	1	E	180 s
192.0.2.0/24	2	B	180 s

RIP ROUTING TABLE

- how far away each destination is — a **distance vector**
- **next hop** information
- RIP's own bookkeeping, such as a **timer** giving the entry its time to live

RIP routing-table entries and timers: RFC 2453 §§3.5, 3.8

RIP'S OWN DATABASE, AND THE TABLE THE DATAGRAMS USE

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RIP routing table at A

destination	metric	next hop	timer
198.51.100.0/24	1	B	180 s
203.0.113.0/24	1	E	180 s
192.0.2.0/24	2	B	180 s

INSTALLS

Forwarding table at A

destination	next hop
198.51.100.0/24	B
203.0.113.0/24	E
192.0.2.0/24	B

IT FEEDS THE FORWARDING TABLE

For each destination, RIP installs its chosen route into A's **forwarding table** — the table A consults for every datagram it must forward.

RIP routing-table entries and timers: RFC 2453 §§3.5, 3.8

RIP'S OWN DATABASE, AND THE TABLE THE DATAGRAMS USE

A RIP router keeps two tables

RIP routing table at A

destination	metric	next hop	timer
198.51.100.0/24	1	B	180 s
203.0.113.0/24	1	E	180 s
192.0.2.0/24	2	B	180 s

INSTALLS

Forwarding table at A

destination	next hop
198.51.100.0/24	B
203.0.113.0/24	E
192.0.2.0/24	B

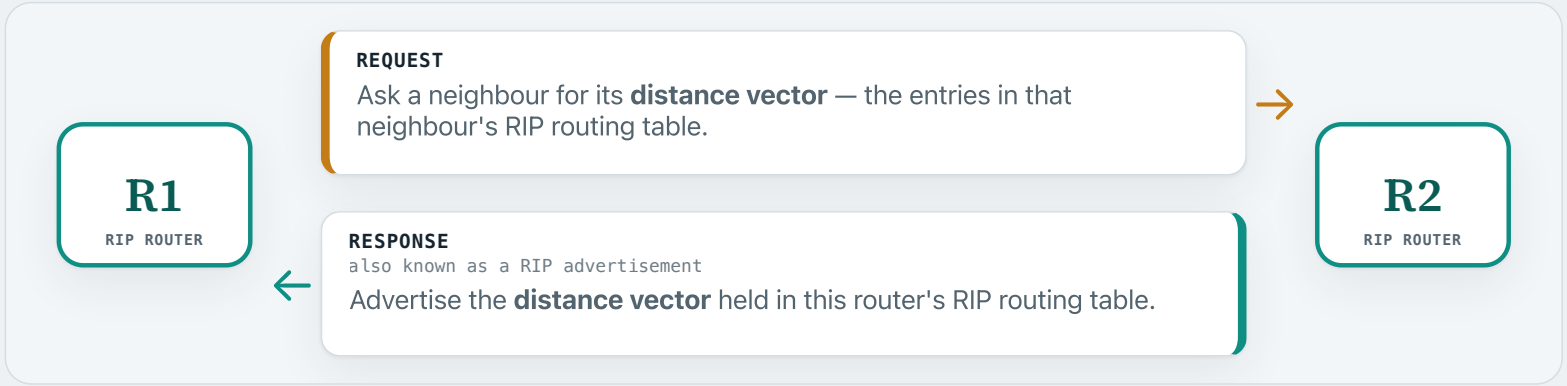
TWO TABLES, NOT ONE

They are kept separately, and may not always agree.
e.g. a route may be removed from the forwarding table, yet still be kept temporarily in the routing table.

RIP routing-table entries and timers: RFC 2453 §§3.5, 3.8

TWO MESSAGE TYPES · THREE REASONS TO SEND A RESPONSE

RIP carries Request and Response messages over UDP



AFTER A REQUEST

Requested

Return all routes or the selected routing-table entries requested by another router.

ROUGHLY EVERY 30 SECONDS

Periodic

Send the current routing table to neighbouring RIP routers.

AFTER A ROUTE CHANGES

Triggered

Send changed routing information shortly after the change occurs.

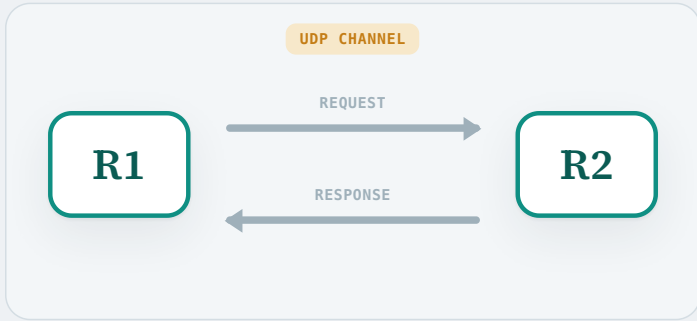
RIP Request and Response processing: RFC 2453 §§3.8, 3.9.1–3.9.2, 3.10–3.10.2

WORKED EXAMPLE · LEARNING, REFRESHING, AND LOSING A ROUTE

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	6	R5	132 s



1 · Can R2 do better?

R1 already reaches `203.0.113.0/24` through next hop **R5** at **cost metric 6**, with **132 s** left to live on the entry. But R1 would like to know whether R2 can offer anything better for this destination.

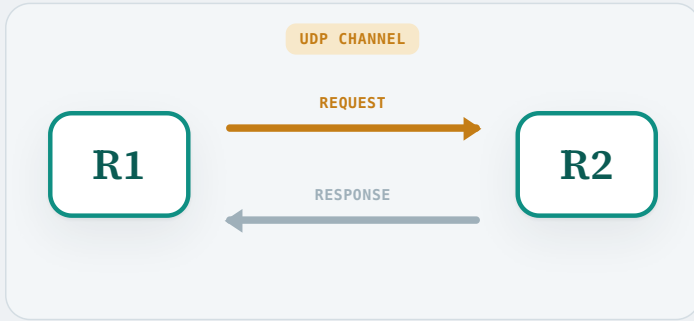
RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

WORKED EXAMPLE · LEARNING, REFRESHING, AND LOSING A ROUTE

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	6	R5	132 s



2 · Ask R2 what it has

R1 sends a RIP **Request** to R2 over UDP, asking for R2's entry for this destination.

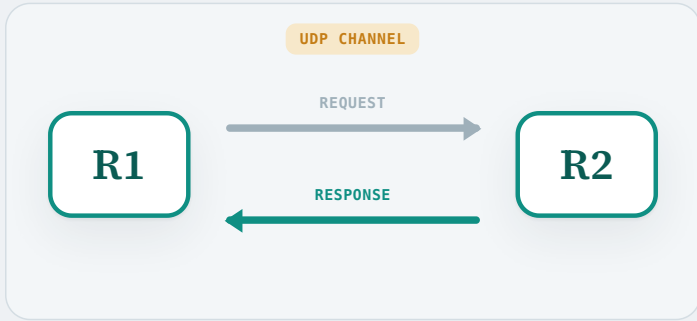
RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

WORKED EXAMPLE · LEARNING, REFRESHING, AND LOSING A ROUTE

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	6	R5	132 s



3 · R2 advertises its distance vector

R2 replies with a **Response**: it can reach `203.0.113.0/24` at **cost metric 2**.

RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

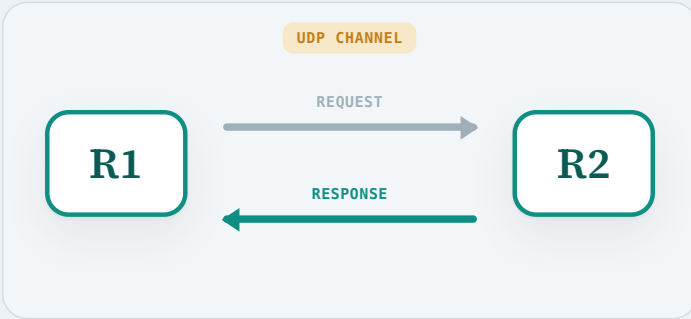
WORKED EXAMPLE · LEARNING, REFRESHING, AND LOSING A ROUTE

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	6	R5	132 s

VIA R2 2 + 1 = **3** < current **6**



4 · Is it any better?

R1 is one hop further along, so going through R2 would cost **2 + 1 = 3**. That beats the current **6**, so R1 takes it.

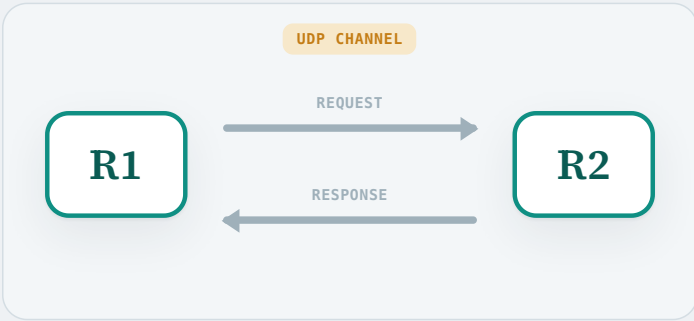
RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	3	R2	180 s

VIA R2 2 + 1 = **3** < current **6**



5 · Replace the entry, then install it

- **Rewrite the row.** Next hop becomes R2, cost metric becomes 3, and the 180-second timer starts again.
- **Install it.** R1 puts the new route into its forwarding table, and starts forwarding datagrams for this prefix to R2 instead of R5.

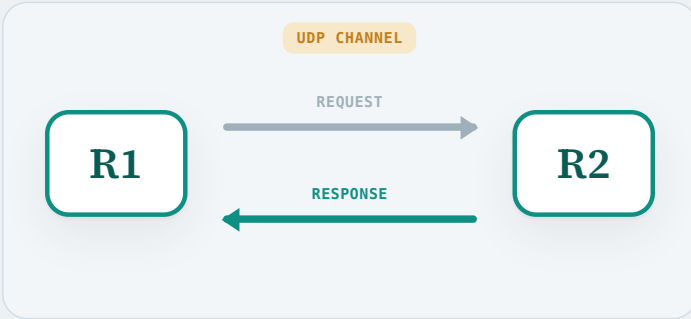
RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

WORKED EXAMPLE · LEARNING, REFRESHING, AND LOSING A ROUTE

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	3	R2	180 s ↻



6 · R2 keeps it fresh

- **Roughly every 30 seconds**, R2 sends a **periodic Response** re-advertising the same route, telling R1 it is still current.
- **Every refresh resets the timer** back to 180 seconds.

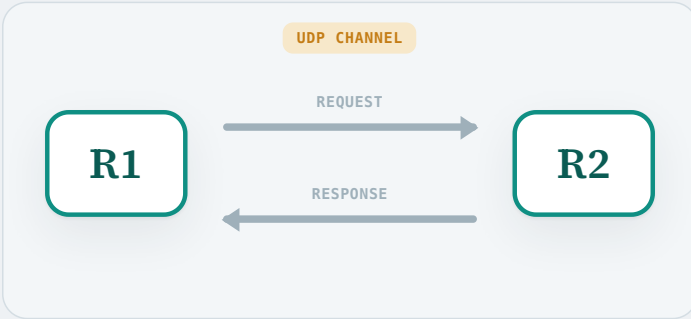
RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

WORKED EXAMPLE · LEARNING, REFRESHING, AND LOSING A ROUTE

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
203.0.113.0/24	3	R2	expired



7 · What if R2 stops sending them?

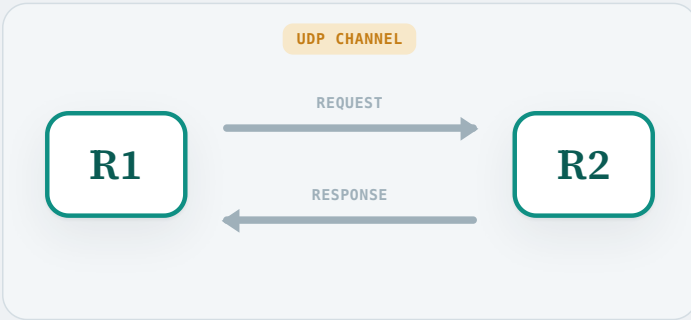
- **The timer runs out.** No valid refresh arrives from R2, and after 180 seconds the entry expires.

RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

An example of running RIP

RIP routing table at R1

destination	metric	next hop	timer
~~203.0.113.0/24~~	~~16~~	~~R2~~	~~120 s~~



7 · What if R2 stops sending them?

- **The timer runs out.** No valid refresh arrives from R2, and after 180 seconds the entry expires.
- **Mark it unreachable.** R1 sets the cost metric to **16** — RIP's infinity — and immediately drops the route from its **forwarding table**.
- **Keep it, to tell the neighbours.** The entry stays in the RIP routing table for another **120 seconds**, advertised with metric 16 so the neighbours learn the destination is gone.
- **Then remove it.** After those 120 seconds, R1 deletes the entry from its routing table altogether.

RIP route maintenance, timeout and garbage collection: RFC 2453 §§3.8, 3.9.2, 3.10

RIP'S PROTECTIONS

How RIP limits count to infinity

> Recall from Module 1: a distance-vector protocol suffers from the **count-to-infinity problem**. RIP guards against it in three ways.

SPLIT HORIZON

Do not send it back

Never advertise a route back to the neighbour you learned it from. If R1's route to a destination came from R2, R1 leaves that destination out of the updates it sends to R2.

POISON REVERSE

Send it back as infinity

Stronger: R1 does advertise the route back to R2, but with a cost of **infinity**.

Both are only **heuristics**. They break a loop between **two** routers, but not one running through three or more.

COUNT ONLY TO 16

Make infinity deliberately small

Therefore, this is why in RIP the maximum path length is bounded to **15 hops**.

Split horizon, poisoned reverse and RIP infinity: RFC 2453 §§3.2, 3.4.2–3.4.4, 3.9.2

INTRODUCING ANOTHER INTRA-AS PROTOCOL

OSPF · Open Shortest Path First

What it is. OSPF is an intra-AS routing protocol which uses the **link-state** routing algorithm.

Protocol channel. OSPF carries its protocol packets directly over IP, without TCP or UDP.

OSPF architecture and packet encapsulation: RFC 2328 §§1, 4.3

QUICK REFRESHER · FROM MODULE 1

A quick recap of the link-state routing protocol

1. Each router creates an **advertisement** for its local information, and tells the others.
2. Every router uses all the advertisements to build the **same topology**.
3. Each router runs a **shortest-path algorithm** over that topology to derive its forwarding table.

OSPF link-state operation: RFC 2328 §§2, 12–16

WHY CHOOSE OSPF?

OSPF provides useful features for a large AS

Authentication

OSPF can authenticate protocol exchanges so routers accept updates only from configured participants.

Operator-defined costs

The operator sets each link's cost, and so decides which paths OSPF prefers.

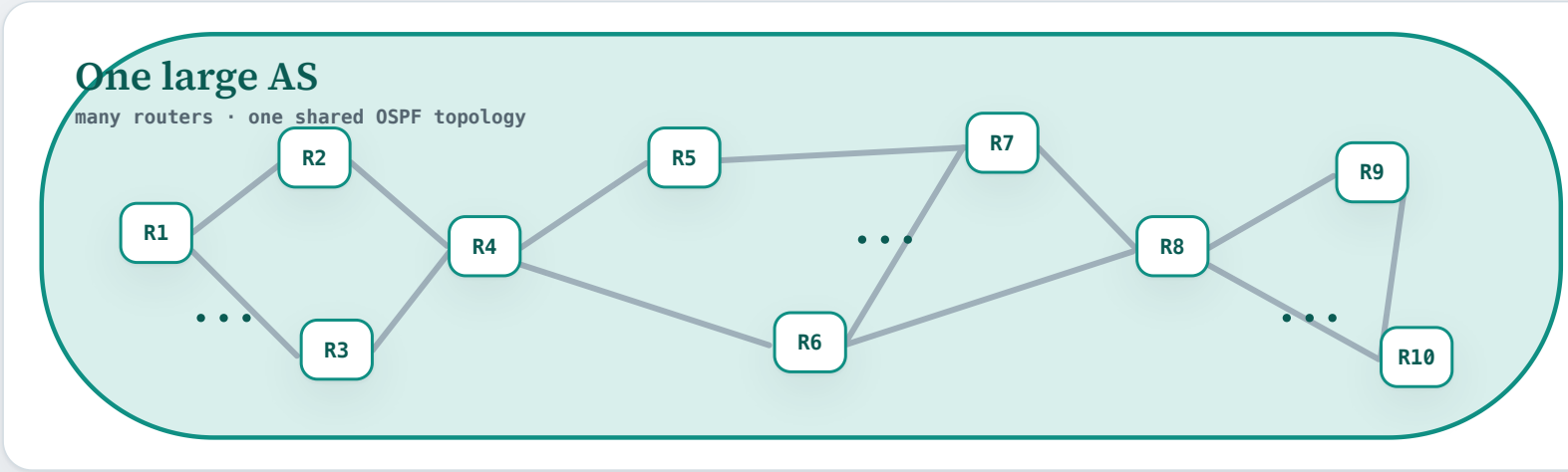
Area routing

A large AS can be divided into **areas**, so that OSPF scales.

OSPF capabilities: RFC 2328 §§2.4, 3, 8 and Appendix D

THE SCALING PROBLEM

Why can a large AS strain OSPF?



More memory

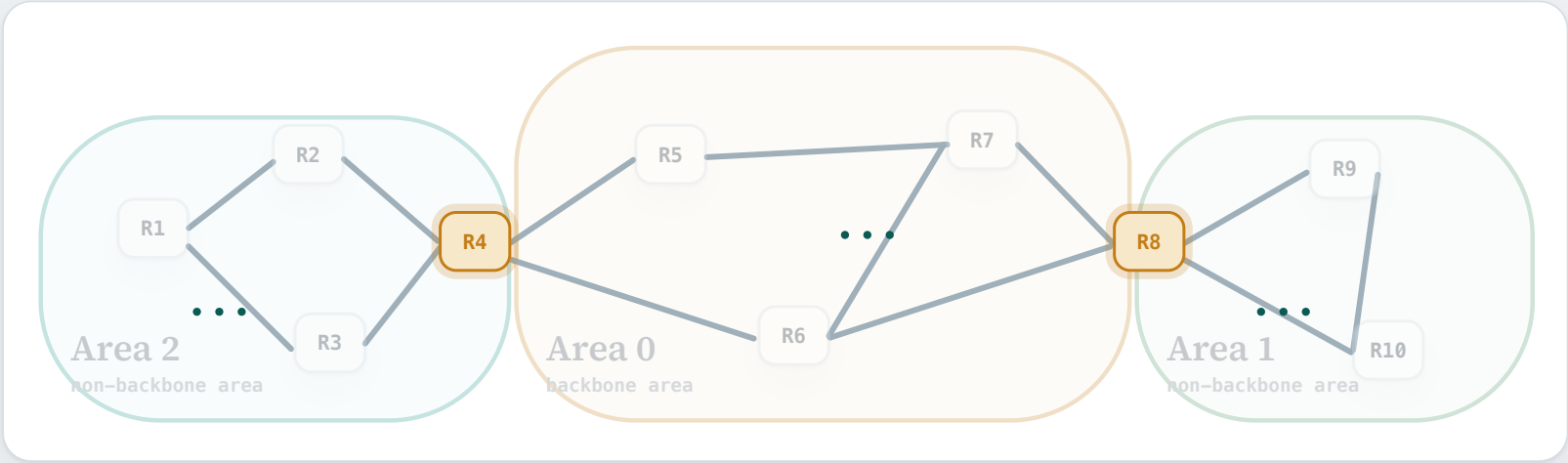
Every router stores the complete AS topology in its link-state database.

Slower convergence

Every change must be flooded widely before all routers calculate the new topology.

OSPF area motivation and topology scope: RFC 2328 §§2.3, 3

OSPF divides one AS into smaller areas



Non-backbone areas

Areas 1 and 2 each contain a smaller group of routers and networks.

Backbone area

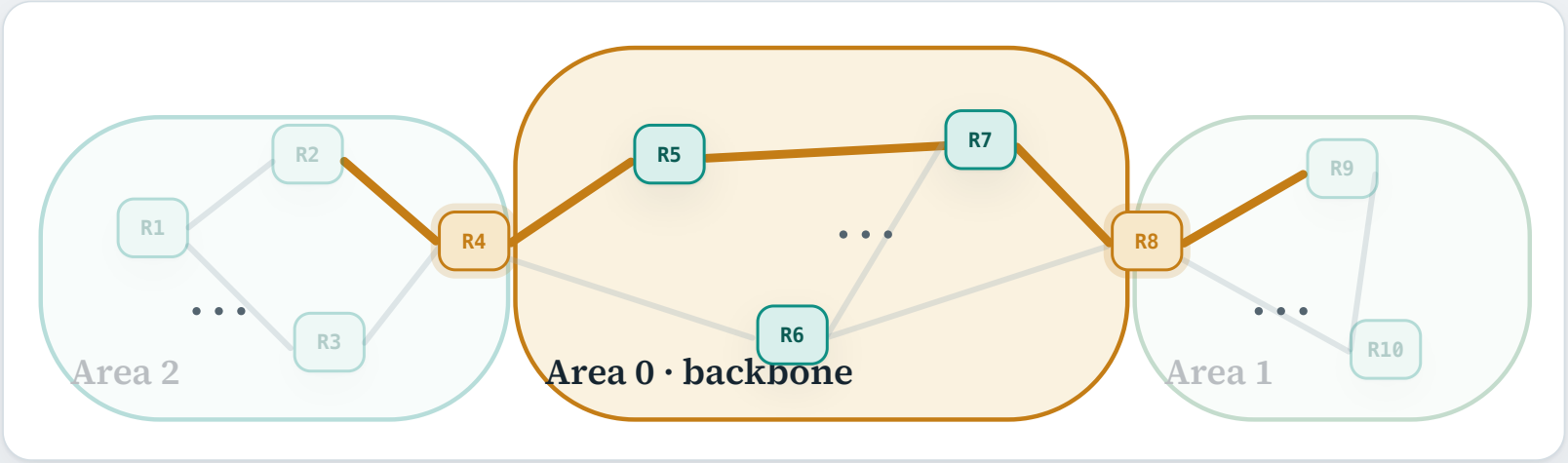
Area 0 connects the non-backbone areas and carries inter-area reachability.

Area border routers

R4 and R8 each participate in Area 0 and one non-backbone area.

OSPF areas and backbone operation: RFC 2328 §§3, 3.1–3.3

How does hierarchical OSPF fit together?



Inside an area

- A router does its link-state flooding only within its own area, and stores that area's topology.
- Each router then runs the shortest-path algorithm over that topology to reach destinations inside its area.

Between areas

- ABRs carry condensed inter-area reachability through the backbone, Area 0.
- For example, a route from Area 2 to Area 1 can follow: `Area 2 → R4 → Area 0 → R8 → Area 1`

OSPF areas and inter-area routing: RFC 2328 §§3, 3.1–3.3, 4.1

Intra-AS routing recap

RIP is distance vector.

- It measures distance in router hops, and caps a path at 15.
- Neighbours exchange distance vectors, kept alive by periodic Responses.

OSPF is link state.

- Routers flood advertisements, so each builds the same topology.
- A large AS is divided into areas so that OSPF scales.

MODULE 2 · KEEPING THE INTERNET ADDRESSABLE

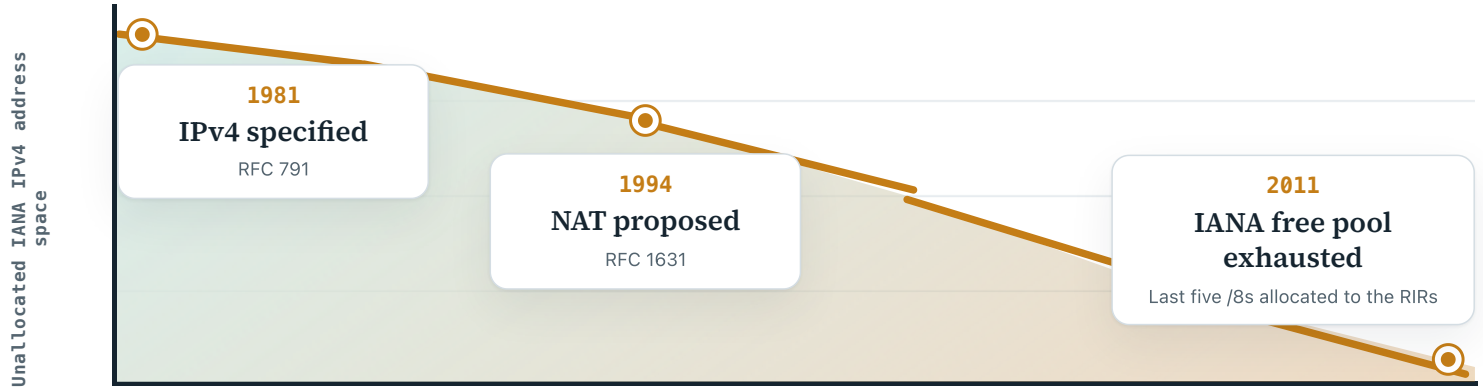
IP Addressing Scaling: Aggregation, NAT and IPv6

1 · Inter-AS routing and AS policy | 2 · Intra-AS routing protocols | **3 · Aggregation, NAT and IPv6**

A BRIEF HISTORY

IPv4's address space became scarce as the Internet grew

2^{32} addresses IPv4 uses 32-bit addresses. Its finite address space became an important limitation as more networks and devices joined the Internet.



RFC Editor: RFCs 791 and 1631 · IANA central-pool exhaustion: ICANN, 3 February 2011

THIS CONCEPT'S ROADMAP

How did the Internet adapt to IPv4 scarcity?

We will study three responses that extended IPv4's useful life and kept the Internet growing.

1

Classless Inter-Domain Routing

already covered in Term 3

2

Share public IPv4 addresses

Network Address Translation (NAT)

3

Use a 128-bit address space

IPv6

A QUICK RECAP FROM TERM 3

CIDR stopped IPv4 address space going to waste

THE OLD PROBLEM

- An IPv4 address is **32 bits**, and before CIDR the split between network and host could fall in only **three places** — after 8, 16 or 24 bits.
- With three sizes to choose from, a great deal of an already scarce space went unused. *e.g.* a site needing 500 addresses had to take a whole **16-bit** block — 65,534 host addresses, nearly all of them wasted.

Classless Inter-Domain Routing: RFC 4632 §§3–4

A QUICK RECAP FROM TERM 3

CIDR stopped IPv4 address space going to waste

20 bits	3	9 bits
the university's block	department	hosts inside that department

department	those 3 bits	its block
Engineering	000	200.23.16.0/23
Science	001	200.23.18.0/23
Law	010	200.23.20.0/23
Arts	011	200.23.22.0/23
…	…	…
Library	111	200.23.30.0/23

THE ANSWER

- Under **CIDR** the split can fall at **any bit**.
- *e.g.* a university takes **200.23.16.0/20** and divides it into eight sub-blocks: **3 bits** to say which department, and the remaining **9 bits** to say which host inside it.

Classless Inter-Domain Routing: RFC 4632 §§3–4

WHAT CIDR COSTS THE ROUTERS

IPv4 address aggregation

network	destination	next hop	in binary
Engineering	**200.23.16.0/23**	uc-gw	**11001000 00010111 0001**0000 00000000
Science	**200.23.18.0/23**	uc-gw	**11001000 00010111 0001**0010 00000000
Law	**200.23.20.0/23**	uc-gw	**11001000 00010111 0001**0100 00000000
Arts	**200.23.22.0/23**	uc-gw	**11001000 00010111 0001**0110 00000000
…	**…**	uc-gw	…
Library	**200.23.30.0/23**	uc-gw	**11001000 00010111 0001**1110 00000000

A NEW PROBLEM

- CIDR uses the address space far more efficiently — but it can create a problem for **router memory**.
- For example, a router carrying traffic towards the university may need to keep **all eight** of these entries, even though every one of them sends traffic to the **same next hop**.
- **How do we solve this?**

Classless Inter-Domain Routing and aggregation: RFC 4632 §§1, 3–4

IPv4 address aggregation

network	destination	next hop	in binary
Engineering	**200.23.16.0/23**	uc-gw	**11001000 00010111 0001**0000 00000000
Science	**200.23.18.0/23**	uc-gw	**11001000 00010111 0001**0010 00000000
Law	**200.23.20.0/23**	uc-gw	**11001000 00010111 0001**0100 00000000
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…	**…**	uc-gw	…
Library	**200.23.30.0/23**	uc-gw	**11001000 00010111 0001**1110 00000000

THE ANSWER

- The answer is **address aggregation**.
- In this example, all eight entries agree on their **first 20 bits** and share a next hop. So the eight can be replaced by one entry, **200.23.16.0/20**.

Classless Inter-Domain Routing and aggregation: RFC 4632 §§1, 3–4

WHAT CIDR COSTS THE ROUTERS

IPv4 address aggregation

network	destination	next hop	in binary
the whole university	`200.23.16.0/20`	uc-gw	`11001000 00010111 0001`0000 00000000

THE ANSWER

- The answer is **address aggregation**.
- In this example, all eight entries agree on their **first 20 bits** and share a next hop. So the eight can be replaced by one entry, **`200.23.16.0/20`**.

Classless Inter-Domain Routing and aggregation: RFC 4632 §§1, 3–4

WHEN ONE NETWORK IS REACHED ANOTHER WAY

What if two entries both match?

network	destination	next hop	match for 200.23.18.5
the whole university	200.23.16.0/20	uc-gw	matches — 20 bits
Science	**200.23.18.0/23**	cloud-gw	**matches — 23 bits**

THE FORWARDING DECISION

- The Science department is now reached through a different provider, so its **next hop is different**. A second entry, **200.23.18.0/23**, is now inserted into the router's forwarding table.
- Now suppose a datagram arrives for **200.23.18.5**. It matches **both** destination prefixes — 20 bits on one, 23 on the other.
- **Which entry should the router use?**

LONGEST-PREFIX MATCH

When more than one entry matches a destination address, forward using the entry with the **longest** prefix.

Here 23 beats 20, so the datagram goes to **cloud-gw**.

Longest-match forwarding: RFC 4632 §5.1 · RFC 1812 §5.2.4.3

MODULE 2 · IP ADDRESSING SCALING

Network Address Translation

A FAMILIAR ADDRESS

Why do so many local networks use addresses like 192.168.1.10?

- **A familiar pattern.** You may see addresses such as `192.168.1.10` on devices at home, in an office, or in a campus lab.
- **Why is there no conflict?** If both your home and a campus lab contain `192.168.1.10`, why does traffic sent at home reach the home device rather than the lab device?

NETWORK ADDRESS TRANSLATION

NAT lets many private hosts share scarce public IPv4 addresses

> The answer is **NAT** — Network Address Translation.

Private IPv4 space: RFC 1918 · Traditional NAT: RFC 3022

NETWORK ADDRESS TRANSLATION

NAT lets many private hosts share scarce public IPv4 addresses

Suppose a host in the home network and a host in the campus network use the **same private address**.

Home network

Inside

A home device uses a reusable private address.

`192.168.1.10`

Campus network

Inside

A campus device can reuse that same private address.

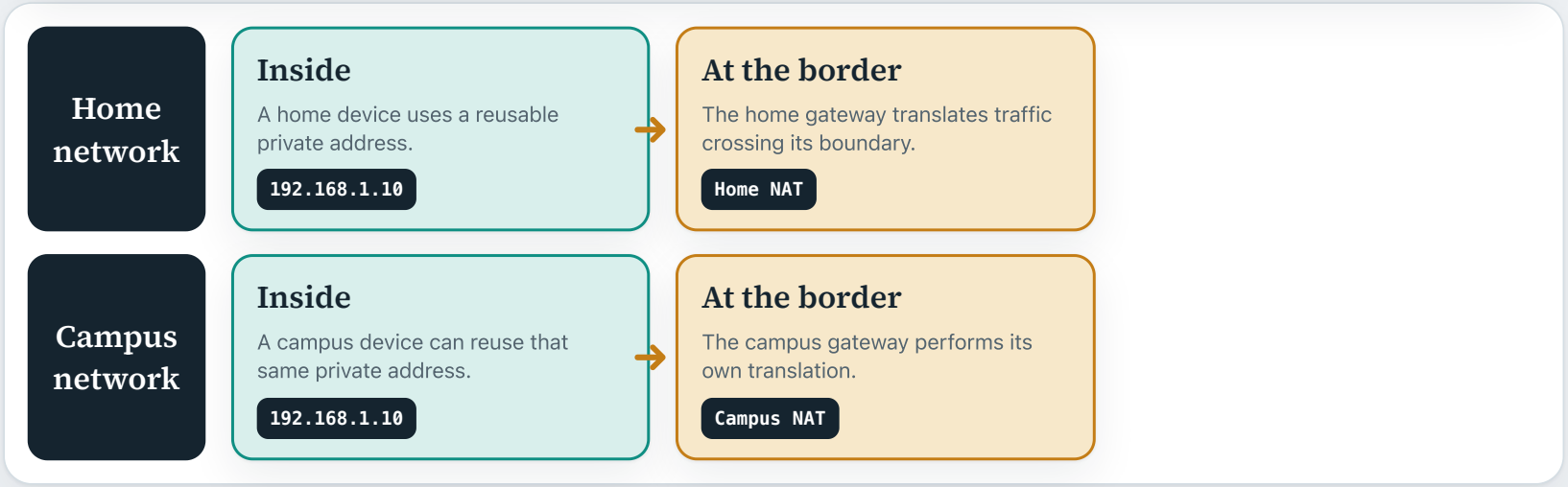
`192.168.1.10`

Private IPv4 space: RFC 1918 · Traditional NAT: RFC 3022

NETWORK ADDRESS TRANSLATION

NAT lets many private hosts share scarce public IPv4 addresses

At each network's border, NAT translates that same private address into **two different public addresses**.

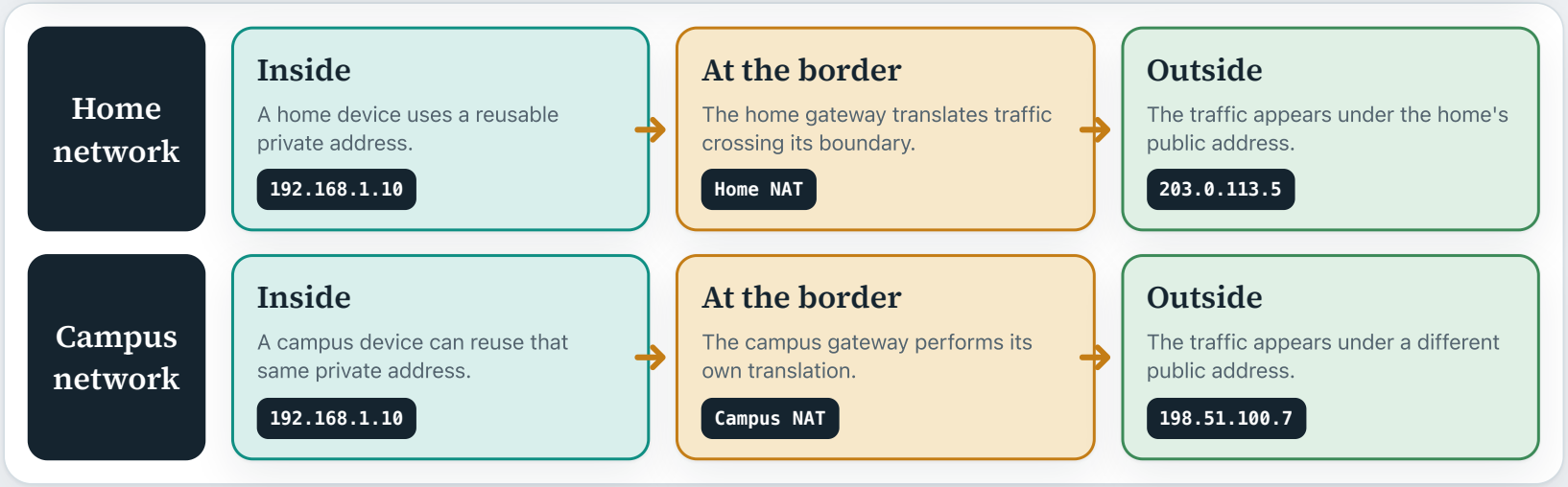


Private IPv4 space: RFC 1918 · Traditional NAT: RFC 3022

NETWORK ADDRESS TRANSLATION

NAT lets many private hosts share scarce public IPv4 addresses

As a result, the outside world only ever observes those two public addresses — and there is **no conflict**.



Private IPv4 space: RFC 1918 · Traditional NAT: RFC 3022

A CLOSER LOOK AT ONE HOME WEB REQUEST

NAT translates both the address and the port

Inside

1 · Private address

Suppose the home device has this private IPv4 address, assigned through **DHCP** (Dynamic Host Configuration Protocol).

`192.168.1.10`

2 · Source port

The device wants to send a request to the server, so it opens a **source port** for this request.

`51514`

3 · Original packet

From that address and port, it sends a packet to the destination server.

```
S 192.168.1.10:51514
D 198.51.100.20:443
```

At the border

4 · Public address

The home gateway has its own unique public IPv4 address, typically assigned by its ISP.

`203.0.113.5`

5 · Assign a public port

NAT chooses an available public port for this outbound connection and saves the mapping.

PRIVATE SOURCE		PUBLIC SOURCE
192.168.1.10:51514	→	203.0.113.5:40001
address : port		address : port

6 · Translated packet

```
S 203.0.113.5:40001
D 198.51.100.20:443
```

Outside

7 · Route to the server

The translated packet is then routed to the destination server.

`198.51.100.20:443`

NAT translation: RFC 3022 §§2.2, 3–4 · Documentation addresses: RFC 5737

THE SERVER'S REPLY

How does the reply reach the correct home device?

NAT return translation: RFC 3022 §§3–4

Advantages and disadvantages of NAT

Advantages

Address conservation: many devices can share one or a few public IPv4 addresses.

Addressing independence: internal addresses can change without affecting the public Internet, and the public address can change without renumbering every internal host.

Disadvantages

Per-flow state: the gateway must remember active translations.

Inbound reachability: internal devices are not directly addressable from the public Internet. A mapping only comes into existence once the device has sent something out, so a reply can find its way back — nothing outside can start the conversation.

Traditional NAT motivations and limitations: RFC 3022 §§1–2, 6

IPv6

- IPv6 is the **newer version** of the Internet Protocol.
- It uses **128-bit addresses**, far more than IPv4's 32-bit format.

IPv6 address architecture: RFC 4291

READING A 128-BIT ADDRESS

IPv6 address

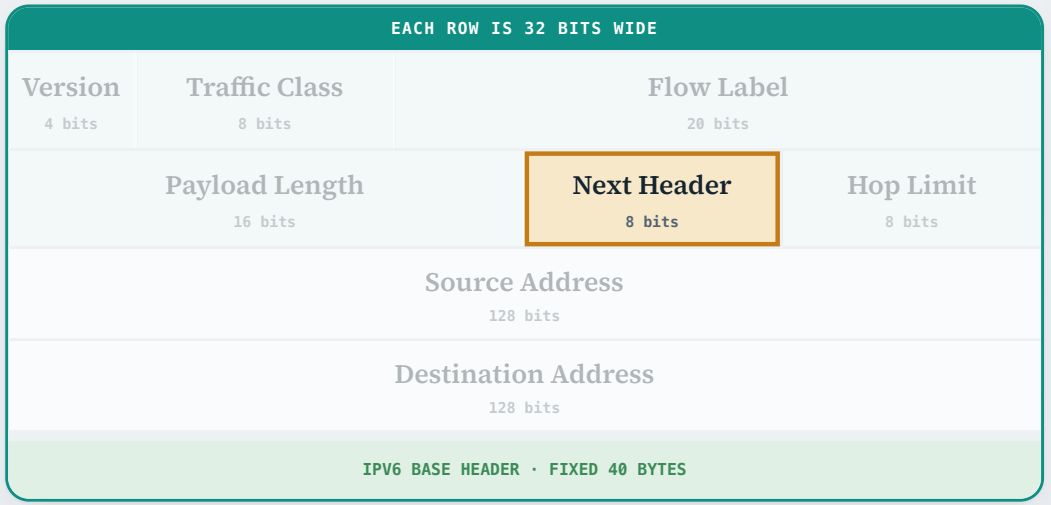
- **First 48 bits · global routing prefix:** identify an allocated site or organization block. For example, a university could be allocated this block.
- **Next 16 bits · subnet ID:** identify one subnet inside that site. For example, this could be the Engineering department's network.
- **Final 64 bits · interface identifier:** identify one interface within that subnet. For example, this could be a laptop's Wi-Fi interface.

IPv6 text representation: RFC 5952 · Address architecture: RFC 4291 · Documentation prefix: RFC 3849

IPV6 HEADER FORMAT

A breakdown of the IPv6 header

Every IPv6 packet begins with the same **fixed 40-byte base header**.



Source and destination addresses

Both fields are 128 bits and identify the packet's endpoints.

Hop Limit

Each router reduces it by one so a packet caught in a forwarding loop is eventually discarded.

Flow Label

Marks packets belonging to the same flow — a TCP connection, for example — so routers can treat them alike.

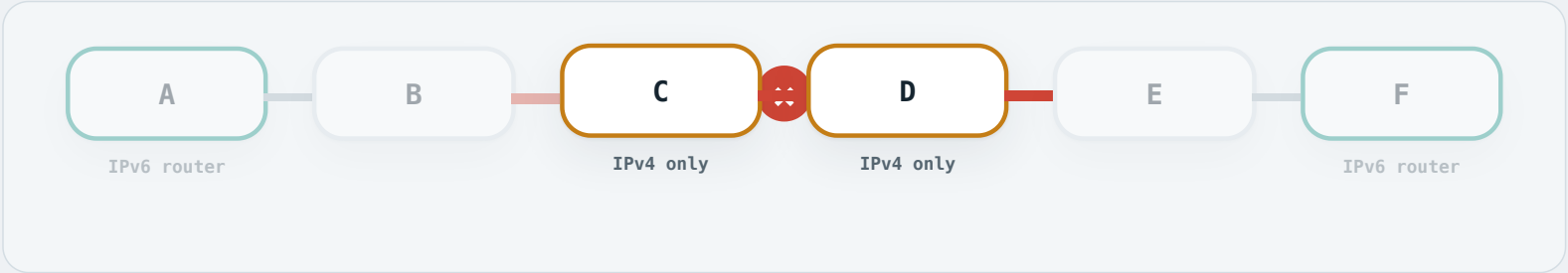
Next Header

Identifies the extension header or upper-layer protocol that follows, such as a TCP or UDP header.

IPv6 base header and extension headers: RFC 8200 §§3–4 · Flow Label: RFC 6437

THE IPV4–IPV6 COEXISTENCE PROBLEM

IPv6 traffic may encounter IPv4-only routers

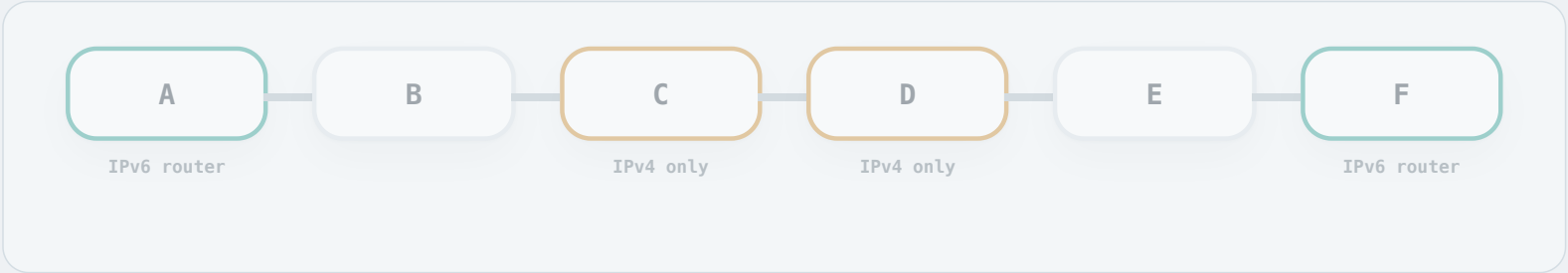


- IPv6 is still being deployed, and progress has been slow: many routers in service are older commercial devices that support only IPv4.
- Suppose IPv6 router A wants to send an IPv6 packet to IPv6 router F.
- The path crosses IPv4-only routers C and D, which cannot forward that packet directly.

IPv4 and IPv6 coexistence during incremental deployment: RFC 4213 §1

THE IPV4–IPV6 COEXISTENCE PROBLEM

IPv6 traffic may encounter IPv4-only routers



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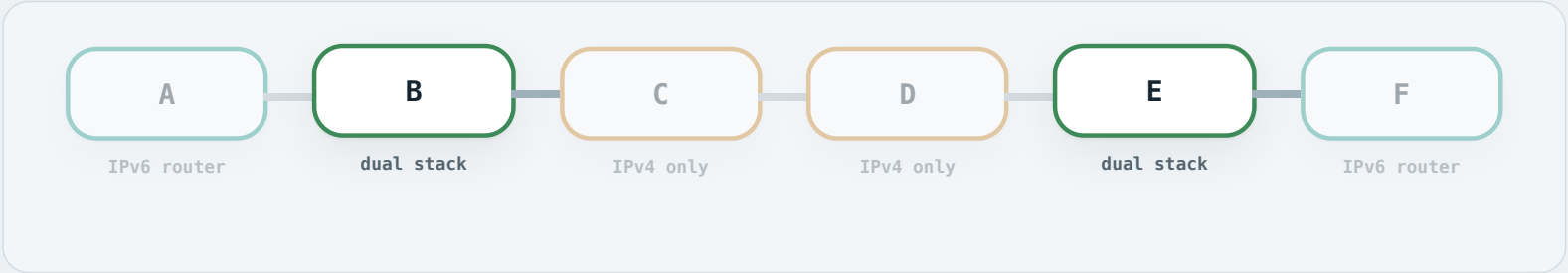
- Suppose IPv6 router A wants to send an IPv6 packet to IPv6 router F.
- The path crosses IPv4-only routers C and D, which cannot forward that packet directly.
- **How can we resolve this compatibility problem?**

IPv4 and IPv6 coexistence during incremental deployment: RFC 4213 §1

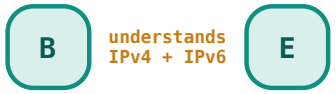
DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region



Dual stack:

A **dual-stack router** runs both the IPv6 and the IPv4 protocol stack, so it understands either kind of packet. Here, suppose B and E are both dual-stack routers.



Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region

Strategy 1 · Translation:

B rewrites the IPv6 header into an IPv4 header. What leaves B is an ordinary IPv4 packet, which C and D forward without trouble.

Fields with no IPv4 equivalent are dropped here.

Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region

Translate back at E:

- When the IPv4 packet reaches E, E rewrites it back into an IPv6 packet and delivers it to F.
- But IPv6 and IPv4 are not equivalent, so this translation is **lossy**: some information is dropped along the way. The IPv6 **Flow Label**, for example, has no IPv4 field to go into.

What was dropped cannot be put back.

Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region

Strategy 2 · Tunnelling:

- B **encapsulates** the entire IPv6 packet — header and payload together — into the **data field** of a new IPv4 packet.

Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region

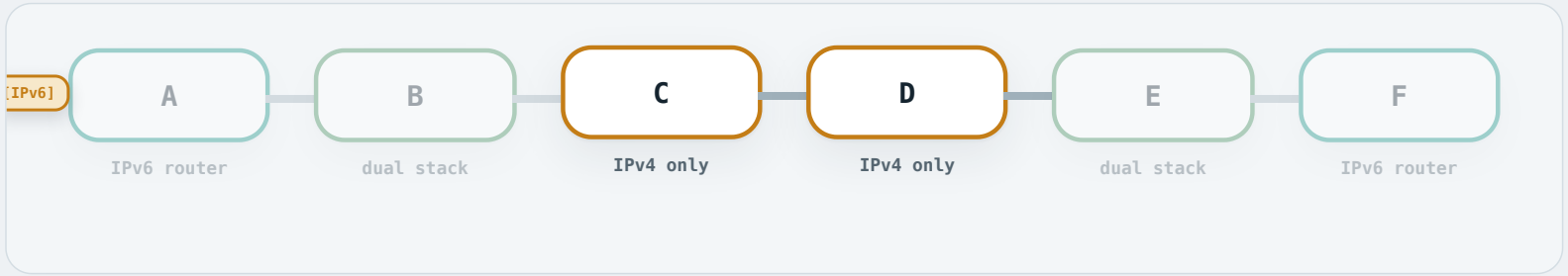
Strategy 2 · Tunnelling:

- B **encapsulates** the entire IPv6 packet — header and payload together — into the **data field** of a new IPv4 packet.
- B then forwards that IPv4 packet to its next hop, C.

Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region



Tunnel:

The encapsulated packet crosses C and D, which see only an ordinary IPv4 packet.

Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region

Decapsulate at E:

E removes the outer IPv4 header, revealing the original IPv6 packet.

Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

DUAL-STACK ROUTERS AT THE BOUNDARY

Two ways across the IPv4-only region

Delivery:

The unchanged IPv6 packet continues to F. Nothing was lost — encapsulation is **lossless**.

Dual IP-layer operation and configured tunnelling: RFC 4213 §§2–3 · Stateless IP/ICMP translation: RFC 7915

THREE CONCEPTS, ONE CONNECTED STORY

Module 2 recap

- **Inter-AS routing:** BGP learns AS-level paths, using eBGP between ASes and iBGP within an AS.
- **Intra-AS routing:** RIP uses distance vector, while OSPF uses link state and areas to route within an AS.
- **IP addressing:** NAT conserves public IPv4 addresses, and IPv6 expands the address space.

BOOKS, STANDARDS, AND CURRENT REGISTRY MATERIAL

References

William Stallings

Data and Computer Communications, 10th edition, Pearson, 2013 · Chapter 19, Routing.

Andrew S. Tanenbaum & David J. Wetherall

Computer Networks, 5th edition, Prentice Hall, 2010 · Chapter 5, The Network Layer.

James F. Kurose & Keith W. Ross

Computer Networking: A Top-Down Approach Featuring the Internet, 3rd edition, Addison Wesley, 2005 · Chapter 4, The Network Layer.

Core Internet standards

RFC 4271, 4632, 5737, 6598, 6888, 8200, and 4213, together with the protocol-specific RFCs cited on individual frames.

Number-resource authorities

IANA Number Resources and APNIC registry, allocation, and address-management material.

Teaching examples

Teaching topologies, addresses, and routing policies are explicitly constructed examples unless a frame identifies an external source.

Standards and registry sources are identified on the frames where they are used.

Acknowledgements

Barry Wu

Provided the current COSC264 slides that formed the content baseline for this deck.

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Developed COSC264 lecture notes used for the routing and addressing material.

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Shared CS340 lecture material used for several Internet routing examples.